

Early Warning Signals and Modelling of Critical Transitions with Applications to Carrying Capacity and Shallow Lake Eutrophication

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Abstract

Suppose you are sitting on a chair and slowly tilting this chair forwards and backwards. At first, you might consider this chair to be quite stable, having the freedom to move in either direction with confidence.

Pushing this, you can tilt slightly further forward or back and still recover and return to your original balance. But once you reach a specific angle (the critical angle) this stability will disappear. Beyond this point, without warning, the chair will flip backwards without control, just a sudden collapse. This fall is irreversible once the equilibrium has been crossed. This analogy is a simpler way to understand how a critical transition works.

Critical Transitions are found in complex systems that occur spontaneously when gradual changes caused by external conditions push a system beyond a tipping point, causing a sudden shift to a different state. In ecosystems this can correspond to cases such as carrying capacity and shallow lake eutrophication. Within this report, we will explore the early warning signals of critical transitions and the mathematical frameworks that model them, specifically focusing on dynamical systems.

There will be a focal point covering fixed points and bifurcations and how we classify them. Using non-linear dynamics we will characterise how these systems behave when reaching a threshold and represent them graphically. In addition, we will explore critical slowing down and increased variability (early warning signals that are specific to ecosystems), with references to stochastic differential equations that react when reaching bifurcation points.

This mathematical framework will then be applied to ecological systems, illustrating how gradual changes in the environment can cause abrupt shifts in ecosystems. By linking the theoretical theory and ecological interpretations, this report will highlight the predictive potential and limitations of early warning signals in ecological settings.

Overall, this report will unify the modelling of critical transitions as well as demonstrating how dynamical system theory can be used to understand the occurrence of abrupt changes in ecosystems.

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1 An Introduction to Critical Transitions and their Early Warning Signals

1.1 Critical Transitions

1.1.1 What is a Critical Transition?

When we observe the world that encompasses us, it is apparent that there is a variety of complex systems that contain several interacting components. Within these complex systems, a diverse range of behaviours can be observed [139]. Among these, there are complex systems that show abrupt shifts from one state to another. These transitions are known as critical transitions and can be seen in many environments and systems (physical or biological) [104]. In general, we can define critical transitions as sudden changes in complex systems that occur when the underlying conditions gradually change and cross a threshold [106]. The critical transitions found within these systems can be described as dynamical processes established by the framework of bifurcation theory, which will be discussed in section 1.3 [139].

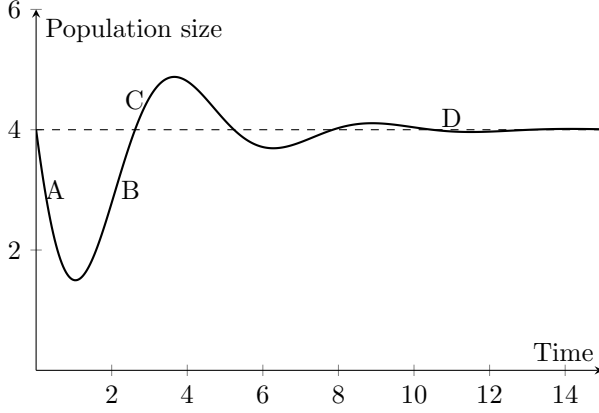
1.1.2 Equilibrium Theorem

At the center of critical transitions are notions of equilibrium and stability. Critical transitions occur when a system moves from one equilibrium to another, caused by a change in parameter [71]. For example, the temperature of the Earth. Earth's temperature is a result of the gain of heat from solar radiation and the loss of heat which is radiated back into the atmosphere. This process results in a stable equilibrium, that is, if there is an increase in temperature, it will return to the original equilibrium [104]. Using equilibrium theory, we can say that critical transitions generally have these attributes:

1. An abrupt qualitative change in the dynamical system occurs.
2. Compared to regular dynamics, these changes occur rapidly.
3. This system will then cross a threshold near a transition.
4. The new state of this system is now far from its previous state [104].

1.1.3 Carrying Capacity example

An example of equilibrium can be found in carrying capacity. Carrying capacity can be defined as the maximum population level that a specific environment can support within a finite amount of resources [34]. Initially, we will assume that our chosen environment is at an equilibrium density, a result of a strong balance between birth and death rates. This environment exhibits a stable equilibrium. If this environment experiences a decrease in population, there will be a higher amount of resources for the organisms that survived which will promote birth rates and decrease death rates, causing the population to increase and the equilibrium density to be restored. Alternatively, if this population thrives beyond the carrying capacity the reverse of this process will occur [104]. This can be represented graphically, as shown:



Carrying Capacity Model

- **A:** Population decreases below equilibrium
- **B:** Rapid increase towards the carrying capacity
- **C:** Overshoot above carrying capacity
- **D:** Stabilisation at equilibrium

At level 4 on the y -axis there is a dashed line representing the carrying capacity. We can observe damped oscillations varying over this line, showing how the population settles over time.

1.1.4 Examples of Critical Transitions

Critical transitions can be seen on a worldwide scale, in many different brackets, for example in medicine there can be spontaneous systematic failures (such as asthma attacks or epileptic seizures) [133], or additionally in finance with the apprehensions of systematic market crashes [67] [88]. Examples of critical transitions in ecosystems include: abrupt changes in Earth systems such as ocean circulation, climate change [76], rangelands, fish or wildlife populations and how they may threaten the landscape of an ecosystem [107] [28]. In section 2 of this report, we will focus specifically on shallow lake eutrophication [108], and apply logic introduced in this section to show the applications of these concepts in a real-world situation.

1.2 Non-linear dynamical systems

1.2.1 An introduction to Dynamical Systems

Critical transitions are a diverse topic covering many corners of mathematics, such as real or complex surfaces, functions or maps, vector fields, differential or integral equations, many of which are examples of dynamical systems [2]. Dynamical systems describe the interactions of processes [104]. There are two focal dynamical systems, these being differential equations and iterated maps (or difference equations) [125]. A differential equation is an equation that contains one or more functions along with its derivatives, where derivatives describe rate of change [66]. Differential equations are only ordinary derivatives, which separates them from partial differential equations that depend on multiple variables [125]. In continuation, iterated maps are an example of a simple model for dynamical systems [27], built from projections of two constraint sets [38]. Throughout this report, there will be a pivotal focus on differential equations rather than iterated maps. Differential equations can be used to model population growth, [18], radioactivity, chemical reactions, diffusion and countless other circumstances [1]. To apply a differential equation, first the problem must be clearly outlined so a differential equation can be derived to model the problem. Then once solving this equation the results can be equated to real world data [18].

Ecosystems can be modelled as dynamical systems, taking the general form:

$$\frac{dx}{dt} = f(x, r) \quad (1)$$

In this case, x will represent a state (for example nutrient level) and r represents a parameter (such as rainfall) [120]. Applying this to our original definition of a critical transition, we can infer that a critical transition occurs when there is a small change in r causing a qualitative change in the structure of the equilibrium.

1.2.2 Stochastic Differential Systems

Stochastic differential equations can be used to predict critical transitions [36]. Throughout this report there will be references and examples using ‘stochastic differential systems’, so we will now introduce these systems. A stochastic differential equation is defined as a differential equation that has random coefficients. These coefficients can be constant or random functions, but are often given with their statistical properties. Following this, the solution of this equation will be a random function, with the main focus of the problem being finding the statistical properties of this function [129]. This uncertain information can then be modelled as a random distribution [65]. Using Ashwin, P. et al.’s 2012 research, a standard stochastic differential equation can be represented as:

$$dX_t = f(X_t, r)dt + \sigma dW_t \quad (2)$$

Where $f(X_t, r)dt$ represents the underlying behaviour of the system, sharing its structure with an ODE: $\frac{dx}{dt} = f(x, r)$, σdW_t describes any random fluctuations that may occur, and W_t is the Brownian motion (the standard model for continuous randomness) [71]. This parallels the definition of a critical transition, with the underlying behaviour showing an equilibrium and the random fluctuations showing the shifts that can occur in a system.

Now we will define Brownian motion using a definition given by J. Michael Harrison in his book ‘Brownian Models of Performance and Control’. Let X be a stochastic process.

A standard Brownian motion or Wiener process can be defined as a stochastic process X having continuous sample paths, stationary independent increments, and $X(t) \sim N(0, t)$. A standard Brownian motion will almost always start at level zero [49].

Definition. *A stochastic process is said to have independent increments if the random variables $X(t_0), X(t_1) - X(t_0), \dots, X(t_n) - X(t_{n-1})$ are independent when $n \geq 1$ and $0 < t_0 < \dots < t_n < \infty$. There will be stationary independent increments if the distribution of $X(t) - X(s)$ depends only on $t - s$.*

1.2.3 Fast-slow Systems example

A critical transition can be naturally-formulated by using fast-slow systems [71]. A fast-slow system can be described by two sets of ODEs [29], a slow varying parameter that drives the system toward a bifurcation (which is defined in section 1.3) and a fast subsystem, inducing a rapid shift [71]. The mathematical framework of fast-slow systems supply an interpretation for why variance and autocorrelation increase in stochastic differential systems before a critical transition occurs [29] (refer to appendix A).

$$x' = f(x, y),$$

$$y' = \epsilon g(x, y) \quad (3)$$

The ODEs in (3) form a fast-slow system, where $x \in \mathbb{R}^m$ are the fast variables and $y \in \mathbb{R}^m$ are the slow variables [71]. The derivation of these equations is found in appendix B An application of fast-slow systems in ecosystems can be represented in a predator-prey model. We can say that the fast variable is the interactions between predator and prey and the slow change is the degradation of the environment. This can lead to a critical transition: the collapse of these populations, if the slow change alters the system’s equilibrium before the population can adapt [132].

1.2.4 Logistic Growth: Carrying Capacity example continued

An example of a fast-slow system is logistic growth represented by the logistic equation. The logistic growth model and other density-dependent models derived from logistic growth can exhibit critical transitions, there will be a gradual change of parameter that leads to an abrupt regime shift [37]. The logistic equation is a growth model that describes population increase ($\frac{dA}{dT}$) as a function of population density (A) and two parameters:

$$\frac{dA}{dT} = rA(1 - \frac{A}{K}) [104] \quad (4)$$

Where r is the relative growth rate and K is the carrying capacity of the environment [104]. The logistic model characterises population growth and development to show that in the early stages the relative growth rate and density independent factors are more important as opposed to an emphasis on carrying capacity and density dependent factors [40]. At lower population densities, the logistic growth model (4) approximates the exponential growth because the limitations of density-dependent factors are minimal. In appendix C there is the derivation originally formed by A. Tsoularis of an enhanced and more accurate version of this model. This model is known as the Verhulst logistic equation or Verhulst-Pearl equation and takes almost the exact same form as equation (4):

$$\frac{dN}{dt} = rN(1 - \frac{N}{K}) \quad (5)$$

Where again, r is intrinsic growth rate and K is the carrying capacity. The key difference is $N(t) = N_0 e^{rt}$ [94]. This enhancement improves the accuracy at early-stage population dynamics, by incorporating an exponential component. This model is particularly useful when the initial growth is rapid or the availability of resources is unconstrained [9].

There are 3 main features of logistic growth:

1. $\lim_{t \rightarrow \infty} N(t) = K$ (the population will always reach carrying capacity).
2. $\frac{1}{N} / \frac{dN}{dt}$ is the relative growth rate and declines at a linear rate when population increases.
3. When the growth rate reaches a maximum (inflection point), N_{inf} is exactly half the carrying capacity, $N_{inf} = \frac{K}{2}$ [127].

We will now represent this graphically, using the models given by A. Tsoularis.

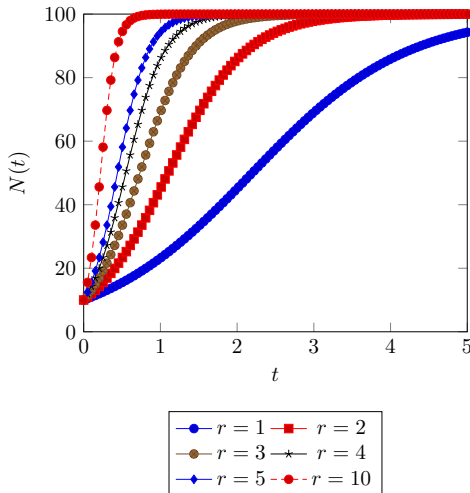


Figure 1: Population growth [127]

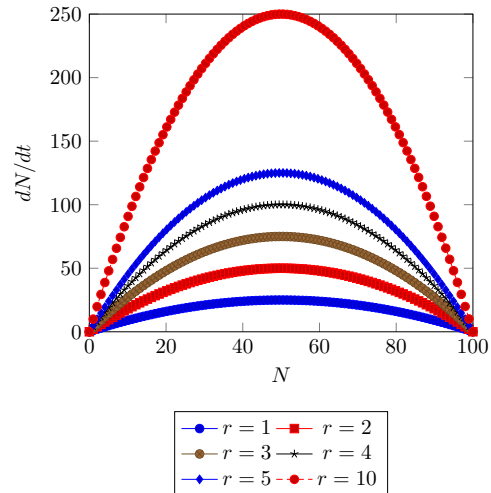


Figure 2: Growth rate [127]

To improve the given models given by A. Tsoularis I changed the given r values, instead I used whole number values of r from 1 to 5 to show how the curve changes as r increases in a set amount, also still using the value of 10 to show a larger jump in figure 2. In figure 3, I instead decided to use the same values of r as figure 2 for the same reasons and for consistency across the two plots. Figure 1 shows the evolution of population size as time increases in terms of the Verhulst logistic growth equation. In this graph we take $N_0 = 10$ and $k = 100$. We can observe that as r increases, the time taken to reach carrying capacity decreases. For figure 3, we take the point of inflection as $N_{inf} = K/2 = 50$ for all values of r [127]. It can be observed that, as r increases the curve is steeper and the maximum growth occurs at the inflection point. This indicates that growth rate depends on the population size, and that the population growth rate increases as the population increases. When $N = K$ population growth stops and then begins to decline, due to factors such as competition and a decrease in resources. This decline is at a larger rate when the population size is bigger. Overall, logistic growth is not constant and is largely dependent on population size, universally peaking at half the carrying capacity [127].

Upon observation of figure 2 we can interpret the nature of the graph as a critical transition. Figure 2 shows a sigmoid curve representing a slow growth, followed by a rapid increase which then becomes saturated [134]. The varying parameters, specifically r and K , cause the system to undergo qualitative changes in behaviour, such as a stable equilibrium to instability or a single steady state transforming into oscillations or chaos, which is shown in bifurcation theory (defined in the next section), which is a core framework to the understanding of critical transitions [120]. Aside from carrying capacity, more examples of critical transitions exhibiting a logistic-growth curve include ecosystem collapse (such as overfishing and desertification), epidemic spread (when resources are limited) and climate tipping elements [104]. To conclude, the logistic growth curve is the simplest visual and mathematical example of how a continuous change in conditions can cause sudden changes (non-linear shifts in the systems behaviour), which builds the fundamentals of the definition of a critical transition [120] [104]. Throughout this report, there will be many references to the logistic growth curve, specifically the Verhulst logistic growth equation.

1.3 Bifurcations

1.3.1 What is a Bifurcation?

To understand what a critical transition is, we next must understand what a bifurcation is. Another way we can define a critical transition is by using bifurcation theory, as a critical transition can be described as a fundamental shift that transpires within a system when it passes a bifurcation [104]. Bifurcations apply dynamical systems with slow varying parameters to transition to far-away attractors [71]. To clarify, the word bifurcation can be used to describe a circumstance in which the qualitative, topological portrayal of the studied subject alters, by a change of the parameters of which the subject is dependent on. A common example of a bifurcation is the ‘catastrophic bifurcation’, that entails once a threshold has been exceeded, in response positive feedback, will back-propel the system, ultimately causing a phase of directional change towards a contrasting state [104]. This shift is known as a tipping point, or a critical threshold, as a sudden shift has taken place causing a contrast to the previous dynamic control [105]. Carrying capacity can be an example of a tipping point, using figures 2 and 3, we can see there is non-linearity, where growth depends on the current state. Following this there are saturation limits, where constraints, such as competition and disease, create feedback. Notably, there is sensitivity near thresholds, where small changes can push the system toward different regimes. These three characteristics build the underlying structure that many real tipping point systems take [104] [76]. Another class of bifurcation occurs when there is a transition from a stable equilibrium to a cyclic or chaotic attractor. How we find bifurcations using fixed points is given in appendix D.

1.3.2 Chaos Theory

Chaos can emerge through sequences of bifurcations, particularly by virtue of the period-doubling route. A period-doubling bifurcation refers to the creation or destruction of a periodic orbit with double the period

of the original orbit [126]. This is Demonstrated (similarly to the previously seen logistic-growth curve) in the logistic map [120] [86]. The discrete logistic map takes the form:

$$x_{t+1} = rx_t(1 - x_t) \quad (6)$$

Where x_t is the current state, x_{t+1} is the next value in time and r is a parameter controlling growth. This logistic map is a canonical example of chaos theory, where small parameter changes can cause an extreme behaviour shift (which is a form of critical transition) [120].

1.3.3 Classification of Fixed Points

Using the eigenvalues given by the jacobian matrix in appendix D, we can classify fixed points defined in appendix D. Fixed points can be modelled by a phase portrait. A phase portrait is the geometrical representation of the trajectories of a dynamical system, providing a description of how a system has evolved over time [7]. Let λ_A and λ_B be eigenvalues of a Jacobian matrix. When both eigenvalues have the same sign, we refer to them as a node. If both eigenvalues are real and $\lambda_A, \lambda_B > 0$, then the fixed points represent an unstable node. Alternatively, if $\lambda_A, \lambda_B < 0$ then we can classify this as a stable node. The behaviour of the nodes is characterised by trajectories that move directly in or out without oscillation. A saddle point occurs when the eigenvalues have opposite signs, $\lambda_A \lambda_B < 0$. This fixed point is always unstable, and exhibits behaviour where trajectories are attracted along one eigenvector direction and repelled along another, forming an ‘X-shaped’ phase portrait. The positive eigenvalue will show an exponential growth and negative eigenvalue will show exponential decay [120]. Now we consider when the Jacobian eigenvalues are complex: $\lambda = \alpha \pm \beta i$. If $\alpha < 0$, then this is known as a stable spiral (or focus). The phase-portrait will show an inward spiral with dampened oscillations. In the case of $\alpha > 0$ there will be an unstable focus, which is represented as an outward spiral [72]. Finally, when $\alpha = 0$, the eigenvalue is entirely imaginary, we will classify this as a centre. Applying this to a phase portrait, this will show closed orbits that neither attract or repel [95].

The following shows the classification of phase portraits graphically for these fixed points, adapted from the qualitative analysis from Strogatz [120] and Boyce [17].

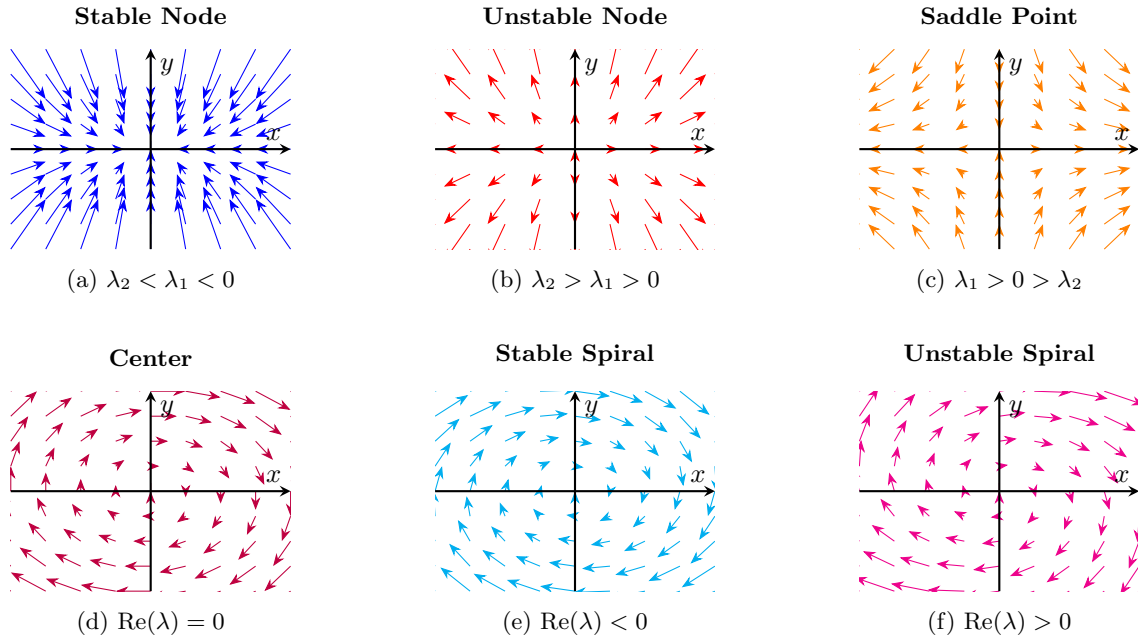


Figure 3: Classification of Phase Portraits for 2D Linear Systems

1.3.4 Classification of Bifurcations

There are four types of bifurcation: fold bifurcation, hopf bifurcation, pitchfork bifurcation, and transcritical bifurcation. We will now explore these four types and show how they can be represented graphically.

Fold Bifurcation A fold bifurcation, also known as a saddle-node bifurcation, describes the collision and disappearance of two equilibria in a dynamical system [74]. This occurs when the critical equilibrium has a zero eigenvalue [74]. This is represented mathematically by

$$\dot{x} = \mu + x^2. \quad (7)$$

Equilibria exist when $\mu < 0$, given by $x = \pm\sqrt{-\mu}$. The lower branch is stable and the upper branch is unstable. The derivation is given in appendix E.1.

Referring back to our logistic growth curve, upon modifying the logistic system it can exhibit many characteristics, such as multiple equilibria and sudden collapse when a threshold is crossed, similar to a fold bifurcation. This mirrors tipping points in ecosystems, a core concept of critical transitions [107].

Hopf Bifurcation A hopf bifurcation occurs when complex conjugate eigenvalues of an equilibrium cross the imaginary axis, this causes the creation or destruction of a periodic orbit (or limit cycle) [120]. The standard mathematical form of a hopf bifurcation is given by [73]:

$$\dot{r} = r(\mu - r^2) \quad (8)$$

If $\mu < 0$ then this is a stable fixed point, if $\mu = 0$ then the system is in critical oscillatory state and if $\mu > 0$ then within the system a stable limit cycle will emerge [120]. The derivation of this is done in appendix E.2

Following this, we have used Strogatz's description to form a diagram for the Hopf Bifurcation. Strogatz states the hopf bifurcation diagram will show the emergence of a limit cycle, of which the amplitude will grow with $r = \sqrt{\mu}$ for $\mu > 0$, replacing the previously stable equilibrium [120].

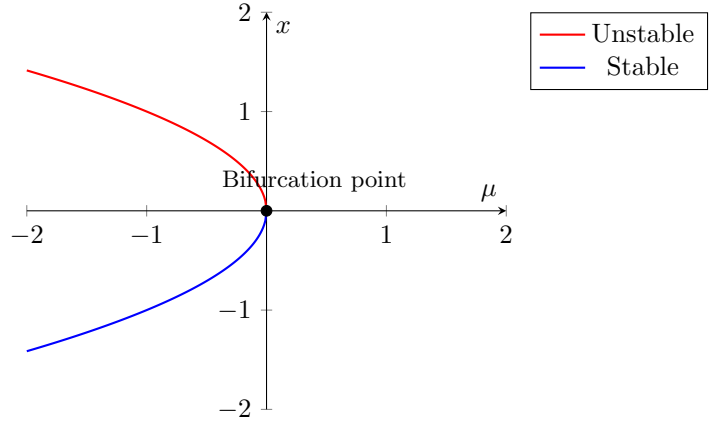


Figure 4: Fold (saddle-node) bifurcation diagram

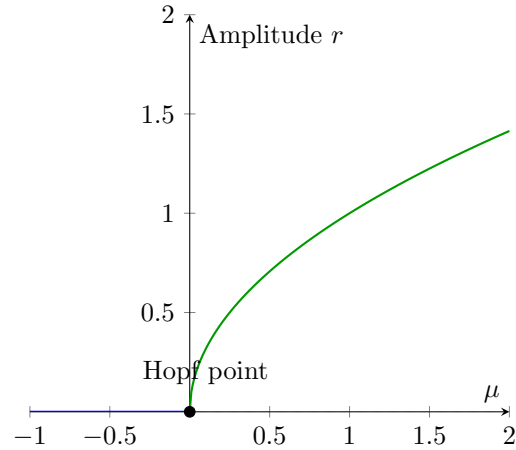


Figure 5: Hopf Bifurcation diagram

Pitchfork Bifurcation Pitchfork bifurcations take their name because their bifurcation diagram looks like a pitchfork [24]. This type of bifurcation is common in problems that have symmetry [120]. In general, a pitchfork bifurcation is a type of bifurcation that occurs when a parameter changes, causing an equilibrium to split into three [25]. A pitchfork bifurcation has the standard normal form:

$$\dot{x} = rx - x^3 [47] \quad (9)$$

This is derived in appendix E.3.

On the right is a bifurcation diagram showing a Pitchfork, formed by the standard normal form as described by Strogatz [120]. As seen in the diagram, at $r < 0$ there is a singular stable equilibrium when $x = 0$, if $r > 0$ then there are three equilibria at $x = 0, \pm\sqrt{r}$ and when $r = 0$ this is where the pitchfork bifurcation occurs (where stability is determined by the cubic nonlinear term [120]).

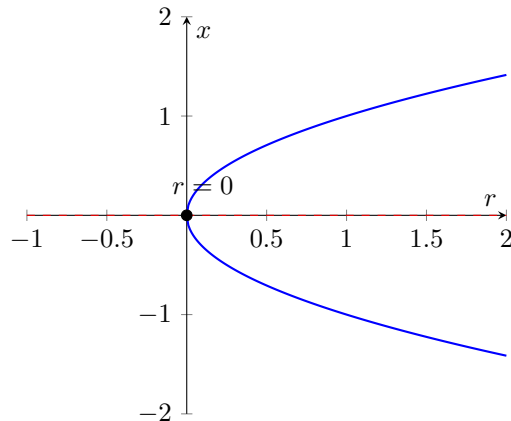


Figure 6: Pitchfork bifurcation diagram

Transcritical Bifurcation A transcritical bifurcation is a type of bifurcation where two equilibrium points intersect and trade their stability as the system's parameter varies [120] [41]. The system takes the normal form:

$$\dot{x} = rx - x^2 [47] \quad (10)$$

Derived in appendix E.4

The diagram on the right shows the graphical representation of the normal form of a transcritical bifurcation formed using the description by Strogatz [120] and Kuznetsov [72].

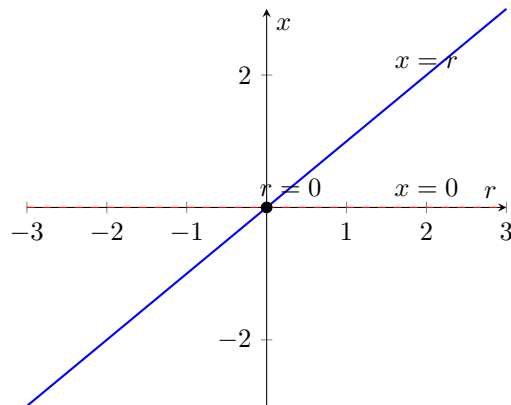


Figure 7: Transcritical bifurcation diagram

1.4 Early warning signs of critical transitions

1.4.1 Examples of EWSs

Early warning signs (abbreviated to EWSs) of critical transitions will be immersed and explored in section 2. Innumerable decisions are made by individuals, industries, and policy makers depend on the accuracy of EWSs for critical transitions [84]. A key reading to properly grasp the EWSs of critical transitions is "Critical Transitions in Nature and Society" by Marten Scheffer. Scheffer suggests that monitoring generic indicators, foremost variance and lag-1 autocorrelation, can determine whether a critical transition is approaching [104]. In appendix A we have already stated the definitions of autocorrelation and variance. We will now explore in more detail the affects of autocorrelation and variance on critical transitions using the descriptions given by Scheffer.

Increasing Autocorrelation Many systems are permanently affected by natural perturbations. In these systems, when approaching a bifurcation, we can expect that slowing will lead to an increase in autocorrelation in the pattern of fluctuations [87]. This slowing causes the rate of change in the system to decrease.

Therefore, lag-1 autocorrelation (the correlation of the time series shifted one time-step back) can be interpreted as slowness of recovery in usual fluctuation patterns. Below is a diagram inspired by figure 1 in the article ‘Early warning of climate tipping points from critical slowing down: comparing methods to improve robustness’, by T.M Lenton [77].

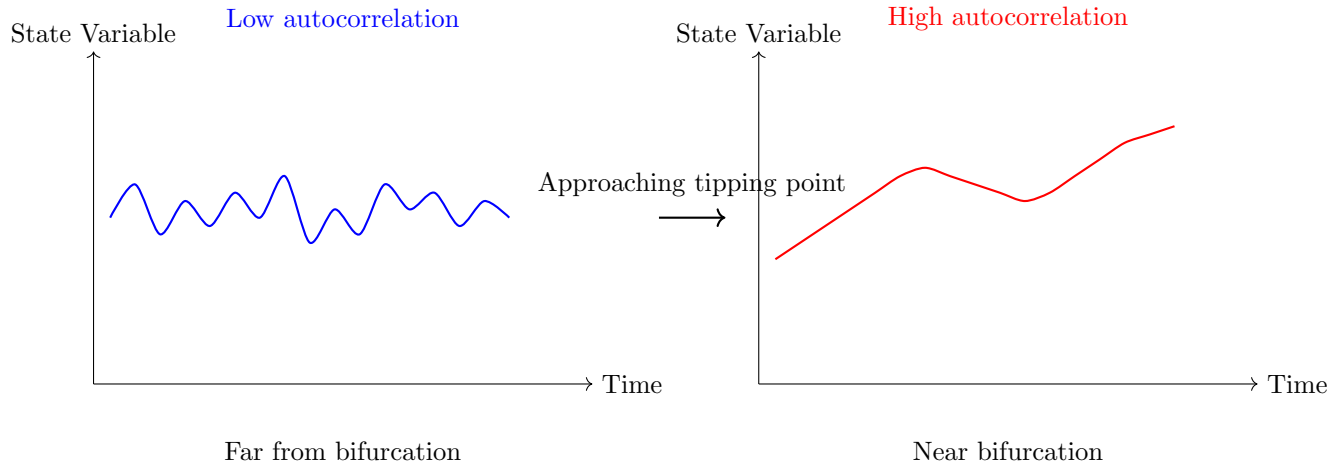


Figure 8: When approaching a catastrophic bifurcation, the recovery from perturbations slows down, which causes following observations to become more similar. Shown by increasing autocorrelation in the time series.

This model is much simpler in comparison to Lenton’s, as Lenton’s uses scientific context focusing on climate tipping point systems and is applied to research [77]. My model shows a more general form using the idea from Scheffer that an increase in autocorrelation is a clear observation when a system is approaching a catastrophic bifurcation where a critical transition occurs [104].

Increasing Variance Fluctuations of Variance can often increase when a critical transition is being approached [22]. When a system is returning back to a stable state close to a transition at a slow rate, this can make the system state drift widely around this stable state [30], this is known as flickering. Both slowing down and flickering cause variance to increase prior to a critical transition [105].

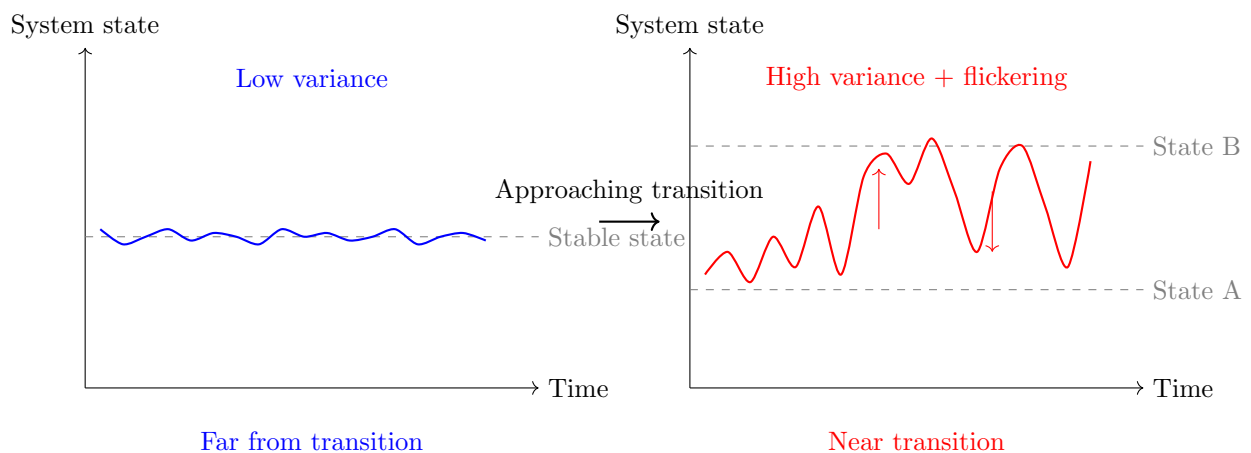


Figure 9: How increased variance and flickering can be used to predict a Critical Transition

Figure 11 was created using concepts from Scheffer [105] and using inspiration from a figure given by Lenton [77]. In comparison to Lenton [77], this figure expands from the framework of EWSs, including flickering and showing a shift in state, using two separate graphs to present time series variance and a slowing recovery.

Critical Slowing Down Referring back to figure 10 during the recovery of perturbations slowing down, represents critical slowing down. A group of EWSs such as increased autocorrelation and flickering have been suggested as tools for predicting critical transitions based on the notion of ‘critical slowing down’ [31]. Systems tend to show critical slowing down as they approach a bifurcation point [120] [104]. Critical slowing down is characterised by an increased return time to equilibrium following a perturbation, which can be caused by an increase in variation [22] and lag-1 autocorrelation [69] [51]. When the rate at which a system recovers from small disturbances becomes increasingly slow, this is known as critical slowing down [106]. This occurs, as an example, when there is a tipping point (also known as the classical fold bifurcation), when a system can be sensitive to a small nudge resulting in a large change [104]. To detect the surging sluggishness of a system, a key identifier can be a reduced rate of recovery, [130] or alternatively can be recognised by a rising ‘memory’ in small fluctuations in the state of a system. This can also be reflected in increased variance [58] and other indicators. For climate and ecology, common EWSs include critical slowing down and increasing variability [106].

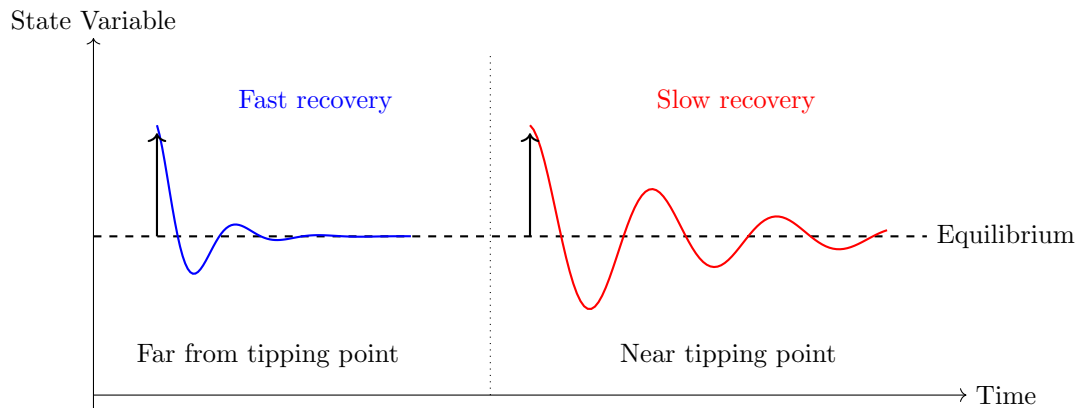


Figure 10: This diagram represents critical slowing down, with blue line showing a rapid return to equilibrium when far from a bifurcation and red line showing a slower recovery as it approaches a tipping point.

This diagram was formed by applying the conceptual illustrations of critical slowing down as described by Scheffer [104]. Figure 12 is explicit in time-series form, where a time series is defined in terms of machine learning as given by Ivan Belvic in the article ‘What is a time series model?’:

Definition. *A time series model is a machine learning model that analyses sequential time series data and can predict future values. Time series datasets consist of data values ordered over time, with time as the independent variable. time series analysis allows for the forecasting of future data values based on previous values in the sequence. [56]*

This use of time-series form differs from that of Scheffer, who implicitly focuses on stable geometry. Instead, this diagram allows direct interpretation of decay dynamics, which more accurately aligns with EWS methods such as lag-1 autocorrelation and variance analysis.

1.4.2 Limitations of EWSs

There are many limitations to these generic EWSs. They can fail to signal a transition if the time series is either too short, too noisy [95], or too non-stationary [26] [3], or if the critical transition is corresponding to a global bifurcation [50] (or in some cases no bifurcation at all [3]) rather than a local bifurcation. It is often assumed that a system contains a saddle-node bifurcation, but can be overlooked in summary-statistic based approaches [12]. For example, variance can be shown to increase for reasons not relating to a transition approaching [114] [113]. In some cases variance won’t increase when a critical transition approaches [81]

[32]. If we restrict our assumption that a system can only produce a saddle-node bifurcation, then using approaches based on concepts such as critical slowing down or increased variance [105] [21] we further assume that a changing parameter has brought this system closer to a bifurcation [12]. This assumption excludes three alternative explanations for the transition in system behaviour:

1. A large perturbation of the system state has moved the system into an alternative basin of attraction [107]. This is an external force that does not arise from the dynamics of the system, so an event that can not be forecast (for instance a sudden increase in fishing effort pushing a harvested population past a threshold) [12].
2. A noise-induced transition. When a chance fluctuation occurs, carrying the system across the boundary [35]. These types of transition cannot be predicted through EWSs, as they are not expected to have the same early warning patterns of increased variance and autocorrelation as seen in the case of a saddle-node bifurcation [81].
3. The system does not pass through a saddle-node bifurcation, but instead gradually moves in a rapid or highly non-linear way, making the detection of any gradual trend unachievable [81].

Overall, we cannot depend on common EWSs when identifying bifurcations and critical transitions, as EWSs are used to indicate whether a system is approaching a tipping point, rather than a tipping point being exceeded [104]. The common indicators mentioned above, increasing autocorrelation, increasing variance, and critical slowing down, represent a decrease in resilience before a critical transition ultimately takes place, but do not produce a time indicator for when the shift has taken place [105]. When applying these signals to real world systems, it can be observed that false positives or false negatives occur. The term 'false positive' refers to the case where an indicator rises but no transition takes place, and 'false negative' refers to a transition occurring without any signals. These conditions occur due to noise, rapid forcing, poor data quality, or incorrect assumptions about the system's dynamics [13]. Therefore, we cannot solely depend on EWSs to recognise whether a critical transition has occurred. To increase reliability EWSs can be combined with structural change analysis, mechanistic modelling, and post-transition evidence [104] [30].

1.4.3 Machine Learning as a Predictor for EWSs

Machine learning techniques can be used for many tasks in the field of non-linear dynamics, including the prediction of critical transitions using the detection of early warning signals [70]. Machine learning models have the ability to automatically capture statistical characteristics, this is done by identifying and learning patterns in data [93], this makes them ideally suited for detecting early warning signals [33]. Machine learning is now widely used in many disciplines including atmospheric science (weather forecasting) [14], economics (finance) [43], biological sciences (medical diagnosis) [78], physical (quantum sciences) [79] and mathematical sciences (non-linear dynamics [93]. However, machine learning tools have not yet been used to classify the most common transitions seen in ecological, financial and climatic systems for both catastrophic (such as first order or discontinuous) and non-catastrophic (such as second order or continuous) transitions [135]. When combating this, machine learners will use training data, but this is reliant upon the real-world systems exhibiting similar properties to one of the bifurcation trajectories within the training set [84]. Ideally, a machine learning classifier would be trained directly on the data of the given real-world system [84]. To avoid the problem of limited data availability, we propose the use of surrogate data methods that will generate trajectories collected from the data of previous critical transitions. Using this, we can generate enough training data to train a machine learning classifier whilst preserving system-specific properties of the data from past critical transitions [11].

Surrogate Data The primary reason for the development of surrogate data methods is to enable robust statistical evaluations about a system from limited data [75] [124] [115]. This is done by creating multiple versions of a time series with the same (or very similar statistical properties), including [75] [10]:

- The **Histogram Distribution** the empirical distribution of a data set obtained by dividing the data into intervals (bins) and counting the number of observations found within each interval. This will show a good representation of how the data is distributed [138].
- The **Power Spectrum** of a stationary random process refers to the distribution of power among the frequency components composing that signal [5].
- The **Autocorrelation Function** of a stationary time series X_t is the correlation between observations separated by lag k , representing the correlation between X_t and X_{t+k} as a function of k [15].

This allows us to test against a null hypothesis using a large number of samples. This is done by using surrogate data methods to boost the number of samples with which to train the machine learning classifiers. Surrogate data is generated from data far from (neutral) and near to (pre-transition) the critical transition. These two data points serve as distinct classes. Following this, machine learning classifiers are trained to distinguish between these two classes. When a classifier assigns a high probability to the pre-transition class, this is interpreted as an EWS for a critical transition [84]. An alternative approach would be to focus solely on surrogate data of past transitions, namely surrogate data-based machine learning (SDML). This approach provides EWSs in empirical and experimental data from subjects such as geology and climatology with higher sensitivity and specificity than two widely used generic EWSs (variance and lag-1 autocorrelation). This approach focuses on surrogates of historical data, therefore it is not bound by the restricting assumption of a local bifurcation like in other methods. This system-specific technique can contribute to improved EWSs to help better prepare for or avoid undesirable critical transitions [83].

2 Shallow Lake Eutrophication: An Example of an Ecological Regime Shift

2.1 Introduction to Shallow Lake Eutrophication with Applications to Critical Transitions

2.1.1 What is Shallow Lake Eutrophication?

Shallow lake eutrophication indicates the process where nutrient enrichment stimulates the growth of phytoplankton (microscopic, photosynthetic organisms that live in aquatic ecosystems such as oceans, lakes, rivers and ponds [98] [39]) to such an extent that dense populations of microscopic suspended algae make the water turbid (cloudy) [104]. In particular, shallow lakes are specifically prone to eutrophication due to morphology (the physical shape and structure of a lake [137]) and human impact [140]. Their physical morphology enhances nutrient recycling and internal feedback mechanisms [111]. Shallow lakes have a small depth which increases sediment-water interactions and reduces dilution capacity, causing them to be more responsive to external nutrient inputs [140]. In general, eutrophication is driven by the elements nitrogen and phosphorus, in the context of the nutrient load of agriculture and wastewater [118].

2.1.2 Shallow Lake Eutrophication as a Critical Transition

A regime shift is a sudden shift in the ecosystem's state [131]. Shifts between alternative states occur in lakes [23] [110]. One of the best studied and most dramatic state shifts is human-induced eutrophication in shallow lakes, observed as a sudden loss of transparency and vegetation [108] [63]. The purest form of most shallow lakes show characteristics such as clear water and rich submerged vegetation. Nutrient loading has changed this situation in many cases [107]. We can define nutrient loading as the input of nitrogen and phosphorus from external sources into aquatic systems [137]. Interestingly, water clarity is barely affected

by increased nutrient concentrations until a critical threshold is passed, shifting lake clarity from clear to turbid [107]. This is our critical transition.

In shallow lakes, either macrophytes or phytoplankton can dominate in the hysteretic range, depending on whether the lake was initially clear or turbid [62] [102] [59]. The phenomenon of 'hysteretic range' is explained by the term hysteresis, meaning two different equilibrium states are possible at identical stressor levels (external drivers that push a system towards a threshold [107]), depending on the initial state [87]. There are two critical stressor levels at which regime shifts can occur, but the direction of change differs per critical stressor level [131]. In lakes, this is associated with a relatively high and low critical nutrient loading. At high critical nutrient loading, regime shifts from macrophytes to phytoplankton dominance can occur. Alternatively, at low critical nutrient loading, macrophytes may start to dominate over phytoplankton [61].

2.1.3 Ecological Feedback Mechanisms for Shallow Lake

Many ecological mechanisms are involved in the process of shallow lake eutrophication, but the entire feedback system is centered around the interactions between submerged vegetation and turbidity. Overall, vegetation has a tendency to enhance water clarity, but in the case of high turbidity there is a prevention in the growth of submerged plants [108]. This is caused by light limitation, this means that submerged plants can only grow down to a certain turbidity-dependent depth from which light availability becomes too low [119]. The positive effect of vegetation on water clarity is the result of a number of different mechanisms:

- Resuspension of bottom material is reduced by vegetation [4];
- Aquatic plants provide a refuge against planktivorous fish for plankton-grazing zooplankton [54];
- Vegetation suppresses algal growth due to a reduction of nutrient availability [128];
- Plants release allelopathic substances that are toxic to algae [89].

Overall, there are two feedback loops, one corresponding to clear-water state and the other to a turbid water-state. In the clear water-state, there are submerged macrophytes that stabilise sediments found at the base of the shallow lake, these macrophytes absorb nutrients and compete with the phytoplankton, this causes the maintenance of water clarity [108] [64]. Alternatively in turbid state, phytoplankton growth is promoted by nutrient enrichment, which in turn reduces light penetration and suppresses submerged vegetation. This causes a loss of macrophytes leading to an increase in sediment disturbance and internal nutrient recycling, reinforcing algae to dominate the lake [108] [118].

Below show a diagram given by Scheffer in his 1993 article: 'Alternative Equilibria in Shallow Lakes' [108] to help visualise the feedback loop.

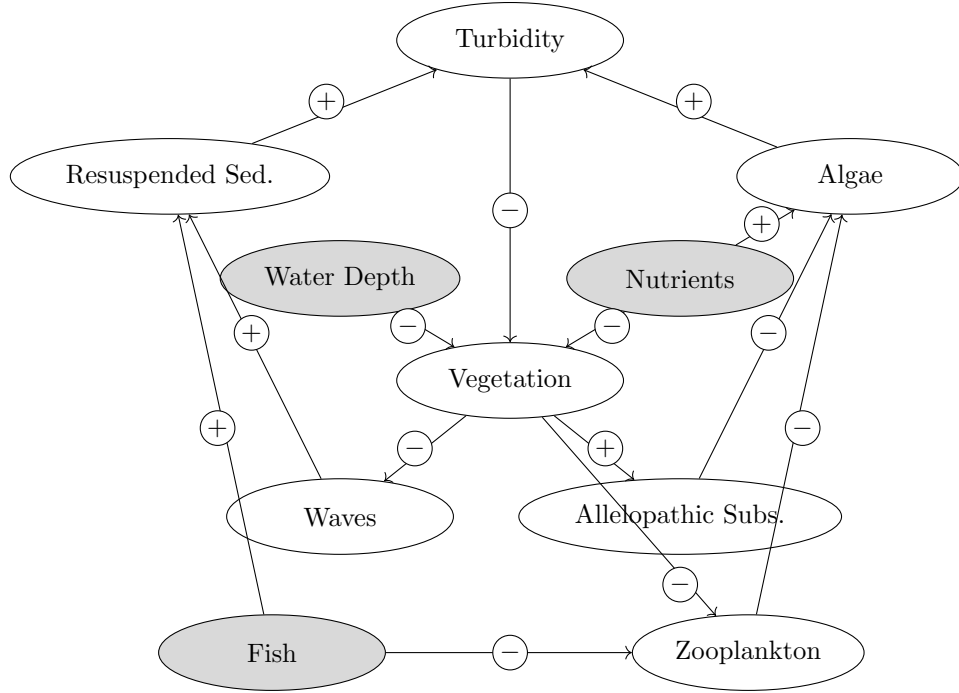


Figure 11: Diagram showing positive (+) and negative (-) interactions between environmental variables and how they influence shallow lake eutrophication. Shaded nodes represent external drivers [108].

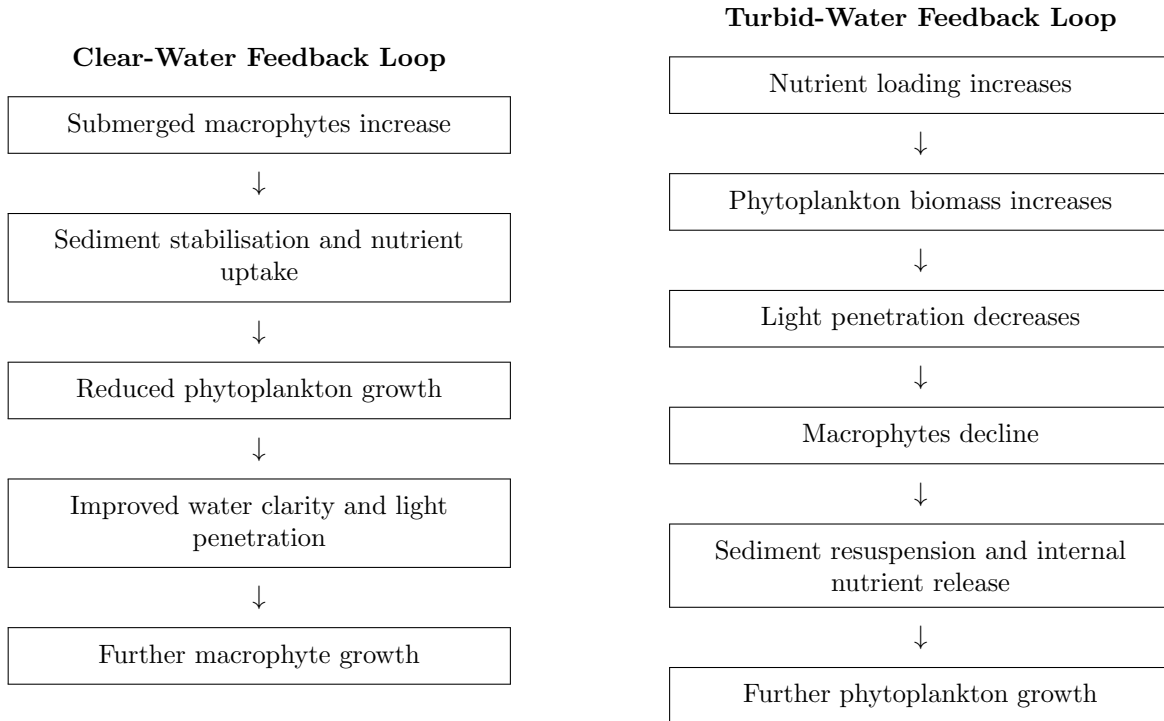


Figure 12: Self-reinforcing ecological feedback loops in shallow lakes. Left: clear-water state maintained by submerged vegetation. Right: turbid-water state maintained by phytoplankton dominance. This diagram was made based on the concept raised by Scheffer [108], which is supported scientifically by Jeppesen [64] and Smith [118], further described below.

2.2 Non-linear Dynamical systems and Bifurcations applied to Shallow Lake Eutrophication

2.2.1 A Mathematical Model Representing Shallow Lake Eutrophication

One way that we can represent eutrophication in shallow lakes is by a simple set of equations given by Scheffer in his book ‘Critical Transitions in Nature and Society’. We will use an inverse Monod function. The Monod function was introduced by mathematician Monod, describing microbial growth rate’s dependence on concentration in a saturating way parallel to enzyme kinetics. The relationship was standardised mathematically, taking the form [90] [117]:

$$\mu s = \mu_{max} \frac{S}{K_S + S} \quad (11)$$

We will observe an inverse Monod function that describes the vegetation effect on equilibrium turbidity denoted as E_{eq} :

$$E_{eq} = E_0 \frac{h_v}{h_v + V} \quad (12)$$

Where E_0 is the turbidity in the absence of vegetation, V is the area of the lake covered with vegetation, and h_v is the vegetation half-saturation constant [104]. To be able to understand the next equation in this set, we must first understand what is meant by the term ‘Hill function’. The Hill function describes a ‘switch-like’ relationship found within biological systems. This is expressed as a sigmoidal response function Θ_n :

$$\Theta_n(E, e) \equiv \frac{1}{1 + (\frac{e}{E})^n} \quad (13)$$

The level e of E required to reach 50% of the maximum response, is known as the threshold value and n refers to the hill exponent [52]. The effect of extinction on vegetation is modelled as a hill function:

$$V_{eq} = \frac{h_e^p}{h_e^p + E^p} \quad (14)$$

In this case $p > 1$ is the hill exponent. Assuming that turbidity and vegetation biomass dynamically approach their equilibrium values in a simple logistic way, equations 25 and 27 can be transformed into the following equilibria [104]:

$$\frac{dE}{dt} = r_E E \left(1 - \frac{E}{E_{eq}}\right) \quad (15)$$

$$\frac{dV}{dt} = r_v V \left(1 - \frac{V}{V_{eq}}\right) \quad (16)$$

In figure 13, we have taken these differential equations and plotted them similarly to figure 7.4 from ‘Critical Transitions in Nature and Society’ by Scheffer [104]. To improve the original plots I changed a few parts to better fit this report. Firstly, I plotted the graphs exactly as the equations are stated, and adding a more detailed scale to better improve the readability. I changed the axis to make it more clear what each graph is referring to. Additionally, i improved the smoothness by adding more plotted points as the Monod and Hill functions require many plotted points to show a smooth curve.

Carpenter formed an essential model for lake eutrophication, stated as:

$$\frac{dP}{dt} = l - sP + \frac{rP^q}{m^q + p^q} \quad (17)$$

Where P is a dynamic variable representing the amount (concentration or mass) in the water column. The rate of P input (measured by mass or concentration per unit time) from the watershed is l . Additionally, s is the rate of P loss per unit time (time^{-1}) [23].

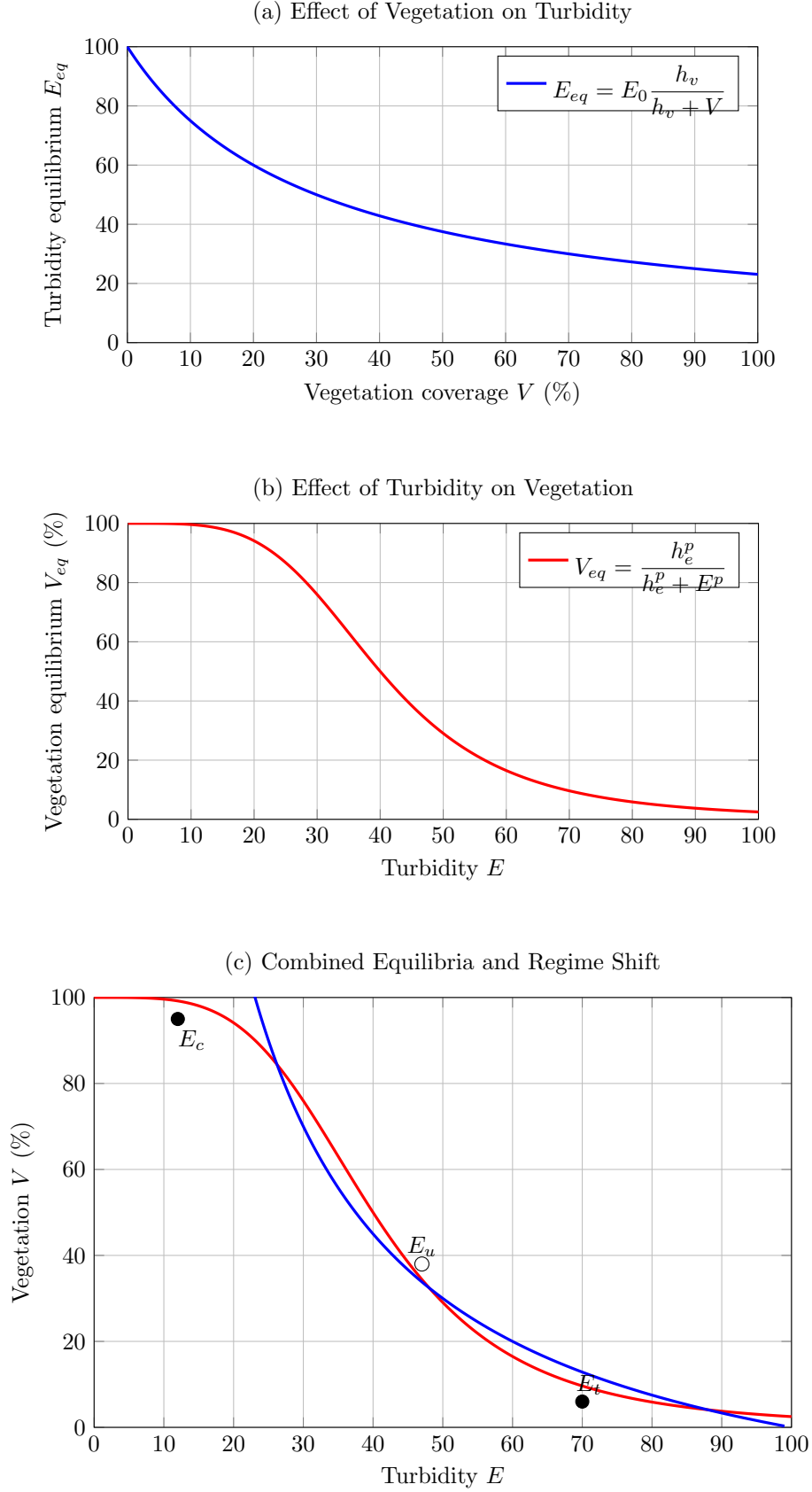


Figure 13: Improved equilibrium diagrams for shallow lake eutrophication model .Blue curve: turbidity equilibrium. Red curve: vegetation equilibrium. Filled circles indicate stable equilibria, open circle unstable equilibrium. The combined equilibrium diagram is meant to represent bistability rather than literal parameter values, hence why the blue lines differ from one another [104].

2.2.2 Interpreting Equilibria and Bifurcation in terms of Shallow Lakes

Appendix F contains the derivation for the classification of the equilibrium type for shallow lake eutrophication, we find that the determinant is

$$\det(J) = r_e r_v [1 - E_0 \frac{h_e}{(h_e + V)^2} \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2}] \quad (18)$$

The equilibrium is stable if $\det(J) > 0$, which means $E_0 \frac{h_e}{(h_e + V)^2} \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2} < 1$. There are three cases to classify the equilibrium:

Case 1 In clear water equilibria, E^* is small and V^* is large. Therefore E^{p-1} is small, and the determinant is small. Considering $\det(J) > 0$ and $\text{trace}(J) < 0$ this will take the form of a stable node.

Case 2 In a turbid equilibrium, E^* is large and V^* is small. This means, $(h_v^p + E^p)^2$ is large, yet again $\det(J) > 0$ and takes the form of a stable node.

Case 3 Intermediate equilibria have intermediate E^* and V^* , causing $E_0 \frac{h_e}{(h_e + V)^2} \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2} > 1$ so $\det(J) < 0$. Therefore this is a saddle point.

E_0 represents the background turbidity, nutrient loading and external pressure on the system [103] [107] [23]. As E_0 increases the clear-water equilibrium moves towards an unstable equilibrium. These equilibrium merge and both disappear [72], implying that the bifurcation type is saddle-node [120] occurring at the bifurcation point $\det(J) = 0$ [72]. Interpreting this ecologically, we can understand that before the threshold two stable equilibria exist, at the threshold the clear-water state vanishes and following the threshold only the turbid state remains [107] [103]. This cannot be easily reversed as reducing E_0 does not immediately restore the clear state because the system is trapped in a turbid basin and must pass a lower threshold to recover [23] [107].

2.2.3 Stochastic Effects of Shallow Lake Eutrophication

Using equation (2) we can form stochastic differential equations that reflect this specific process:

$$dE = f(E, V)dt + \sigma_e dW_t \quad (19)$$

$$dV = g(E, V)dt + \sigma_v dW_t \quad (20)$$

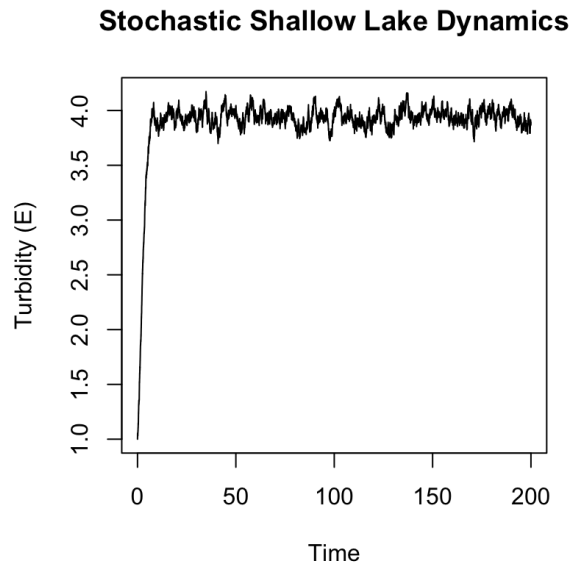
Where $f(E, V) = r_e(E_0 \frac{h_e}{h_e + V} - E)$ and $g(E, V) = r_v(\frac{h_v^p}{h_v^p + E^p} - V)$. The equilibria of this model correspond to ecological states, which provides the mathematical foundation for analysing their stability, as represented by the ecological feedback loop (figures 11 and 12) [108]. Using R we will perform a stochastic simulation of the system. We start by redefining our equations:

$$dE = f(E, V)dt + \sigma dW_t \quad (21)$$

$$dV = g(E, V)dt \quad (22)$$

We have simplified the second equation by removing the noise parameter. Environmental noise is typically encompassed into dynamical models by making one or more model parameters stochastic, and as a result more responsive to environmental disturbances. However, it is unclear whether including noise makes a difference to the model performance [91]. Therefore, we will apply noise to the turbidity variable as a modelling assumption, allowing the effects to be observed in a regulated approach. In our R code, we want to see behaviour such as bistability and tipping, to do this we need to choose specific numbers as our illustrative parameters.

- $E_0 = 4$: This parameter represents external pressure and acts as a control parameter [104]. We have chosen 4 specifically as it is large enough to allow a turbid state, but small enough to allow the existence of more than one equilibrium.
- $h_e = 1$: h_e is the vegetation half-constant and shows how strongly vegetation reduces turbulence [52]. I specifically chose 1 to set the midpoint of the response.
- $h_v = 1$: Representing the turbidity threshold for vegetation collapse, h_v shows the level of turbidity where vegetation drops significantly [64]. 1 is used again for the same reason, helping to maintain symmetry.
- $p = 3$: The hill coefficient shows nonlinearity [52]. When $p = 1$ there is no bistability, when $p > 1$ then a non-linear threshold can be reached. $p = 3$ is a good choice to ensure bistability.
- $r_v = r_e = 1$ These parameters affect the speed of which the equilibrium is reached. 1 is used as it does not affect the equilibria.
- $\sigma = 0.1$ This parameter represents the magnitude of the environmental variability [104]. 0.1 was chosen as it is small enough to not dominate the dynamics but is large enough to be able to observe variance and autocorrelation (key EWSs).



Appendix G contains the R code used to form the graph on the left.

There is a rapid shift between $t = 0$ and $t \approx 10$, where the turbidity increases from 0 to approximately 4. Following this there are stochastic fluctuations around a stable state, which validates the theory of alternate stable states in shallow lakes [107]. The system then remains in a turbid state. This is a rapid adjustment, not a gradual change, therefore no clear evidence to show that it is a critical transition. As we can see, there is no clear indication of increasing variance or autocorrelation preceding the transition, therefore there is a lack of common EWSs that are associated with a critical transition [105]. In section 2.3 we will give more figures using R to show whether shallow lake eutrophication exhibits any EWSs.

Figure 14: Time series of lake turbidity

2.2.4 Fast-Slow systems with Applications to Shallow Lakes

We can interpret shallow lake eutrophication as a fast-slow system. As mentioned previously, fast-slow systems are a natural descriptor of critical transitions. In these systems, there are different variables evolving at diverse speeds. Tipping occurs when a slow variable pushes the system through a bifurcation, hence causing the critical transition [71]. In ecosystems this can be represented by long periods of gradual change, interrupted by regime shifts [109]. Critical points can be observed when shifts occur [111]. In the context of shallow lakes, we can identify the slow change as nutrient loading, with the system showing non-linear abrupt switching as a response, and the fast changing parameter as turbidity [60].

2.2.5 Chaos Theory in Shallow Lake Eutrophication

As stated earlier, chaos refers to dynamics that show sensitive dependence on initial conditions and irregular long-term behaviour [120]. Complex dynamics can be observed in non-linear ecological systems. These dynamics include oscillations and chaotic behaviour, especially in systems with feedback systems [50]. We've already established the feedback systems in shallow lake eutrophication in figures 11 and 12. Studies have been completed on plankton communities demonstrating that complex and chaotic dynamics emerge from non-linear species interactions [6]. Shallow lakes are typically characterised by bistability and regime shifts as opposed to chaos, strong non-linear feedback systems present in eutrophication suggest that complex dynamics can arise under specific conditions [105].

2.3 Early Warning Signals of Shallow Lake Eutrophication

2.3.1 Examples of EWSs for Eutrophication

EWSs are statistical indicators used to predict if a critical transition is approaching [99]. We have already explored how variance, autocorrelation, critical slowing down and flickering can be used to predict critical transitions in ecosystems, we will now apply each of these to shallow lake eutrophication to see if they are observed in this case as well as using R code given in appendix G to form models to see if this works for our SDEs.

In the article 'Flickering gives early warning signals of a critical transition to a eutrophic lake state', there is a case study exploring a multi-decadal time series from the historical changes in the Erhai Lake-catchment system in Yunnan, China, recorded from the 1950s to 2009. From 1980 there is a continuous increase in variance, while skewness and autocorrelation decrease. As a result of sensitivity tests, these findings can be considered reliable [136]. Increased variance cannot be explained by critical slowing down of a system close to an equilibrium due to the decline in skewness and autocorrelation. There is strong evidence eluding towards the presence of exogenous drivers, therefore we can reject alternative explanations for increasing variance based on concepts of internal noise generated by endogenic changes [35]. Where exogenous refers to a variable with a value determined outside the model and not explained by the system itself. In contrast, the value of an endogenous variable is determined within the model as part of the system of equations [116]. Furthermore, the rising variance is most likely a representation of the interactions between multiple exogenous drivers and the crossing of numerous internal thresholds that magnify the system's responses and induce flickering between alternate attractors [19] [22]. In large data sets, the detection of flickering can be formed by distinct probability distributions of system states given by potential analysis [80] [53]. The data for the Erhai Lake is too sparse to perform a full potential analysis, but evidence of flickering in this specific lake-system can be observed in the form of eutrophication events and algal blooms between 1980 and 2000 and in the increased variance [136]. The system dynamics changed over a multi-decadal time-frame. In 2001 a critical transition occurred, foreshadowed by signals of rising variance caused by flickering that occurred ten to thirty years prior. In comparison to mathematical models of lake regime shifts, the eutrophication of the Erhai Lake showed a similar timescale [22] [8].

Another way we can detect critical transitions in lakes is by use of spatial early warning signals (SEWSs). SEWSs are indicators based on changes in spatial characteristics of a system and can provide early warnings of approaching transitions, and can be applied to ecosystems [68]. In an experimental lake study, researchers induced eutrophication and before the transition there was an increase in spatial variance and spatial autocorrelation [20]. This is direct evidence that proves the existence of SEWSs in shallow lakes.

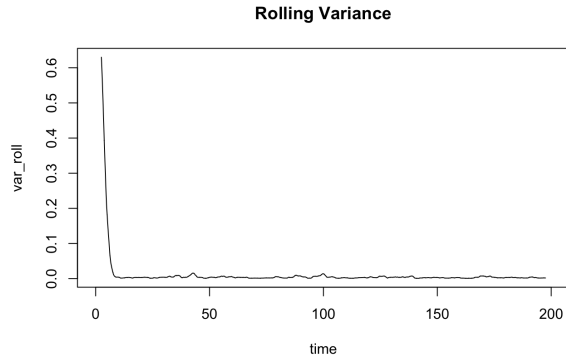


Figure 15: Variance of Turbidity

We have just established using a case study that variance shows a general increase as approaching a critical transition. The diagram on the left was formed using R and the simplified set of equations given in section 2.2.2. Upon observation, it is extremely clear that as the timescale increases, variance decreases. This opposes the arguments that have been given for the EWSs of critical transitions in general and in the specific case of shallow lake eutrophication.

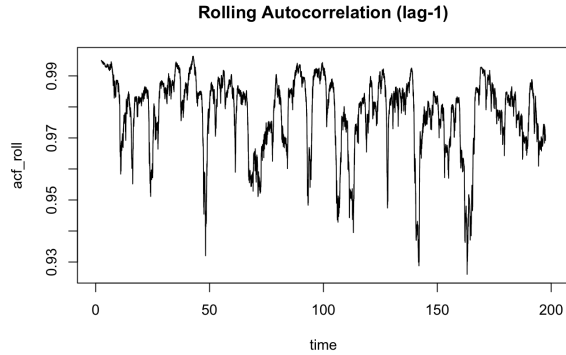


Figure 16: Lag-1 autocorrelation of Turbidity

Referring back to the case study on Erhai Lake, we have stated that there is an autocorrelation decrease. The figure on the right was formed in the same way as figure 15 but instead is showing lag-1 autocorrelation. As we can see there is little to no pattern in the autocorrelation as the timescale increases. There may be signs of a slight decrease, but the large range of values show no direct pattern. This opposes the common EWSs of critical transitions, and suggests that a critical transition has not occurred.

2.3.2 Limitations of EWSs Specific to Shallow Lake Eutrophication

As seen in figures 15 and 16, the common EWSs of increased variance and autocorrelation are not represented. It is easy to jump to the conclusion that no critical transition has occurred, which could be the case. But if the shallow lake has shown an abrupt shift from clear water to turbid then surely a critical transition has transpired [108]. Similarly, the given case study also differs from our general indicators of EWSs for critical transitions, showing a decrease in autocorrelation as opposed to an increases. Nevertheless, we know that a critical transition did occur in Erhai Lake in 2001. Scheffer provides some reasons as to why identifying a critical transition in lake ecosystems can be challenging, which could reason with why both the SDEs graph and case study differ from the norm. There is not an easy to access the nutrient level of the lake as phosphorus is stored in the sediment, making it hard to pinpoint when the critical transition has taken place. Additionally, there is not a single nutrient level for maintaining clear water and different lakes may vary significantly in the nutrient level they can tolerate before they flip to a turbid condition. Size can also affect the eutrophication of shallow lakes, with smaller lakes having a higher chance to be in a vegetated clear state [105]. Overall, collecting information on lake eutrophication can be challenging and often common EWSs won't align with the occurrence of a critical transition, therefore we should not depend solely on EWSs to detect whether a critical transition will occur.

2.3.3 Applying Machine Learning to Shallow Lakes

In shallow lakes, eutrophication is tracked through nutrients (nitrogen and phosphorus), algal biomass and chlorophyll-a [45] [122]. There have been many cases where machine learning has been used to forecast shallow lake eutrophication:

- **Chlorophyll-a and algal biomass:** Machine learning has been used to form tree-based and neural models that can predict chlorophyll-a or algal biomass from local data in eutrophic shallow lakes such as Taihu and Poyang. Neural networks are an example of a machine learning model used. This confirms the strong predictability based on water quality variables [121] [55] [42].
- **Trophic state and risk of bloom** Machine learning approaches have been used to model trophic state indices and bloom risk classifications in lakes. Methods such as artificial neural networks and deep learning models were used to tackle this specific topic. These methods apply key water quality variables such as chlorophyll-a concentration and nutrient levels to classify trophic conditions and predict bloom [57] [97].
- **Nutrient status** Machine learning methods have been applied to predict nutrient concentrations in aquatic systems, such as total nitrogen, total phosphorus and nitrate, using environmental and water quality data [85] [48]. Using these predictors, nutrient ratios can be estimated for nutrients such as total phosphorus or total nitrogen, which are critical in establishing nutrient limitation and phytoplankton growth in eutrophic lakes [112].

Therefore, machine learning is a valuable tool for predicting critical transitions and comprehending the factors of eutrophication in lakes. These techniques help to overcome the limitations of EWSs, and therefore give more accurate predictions for when a critical transition is about to occur, meaning in catastrophic cases machine learning can be used to prevent a shift [96] [33].

3 Conclusion

Throughout this report we have seen how critical transitions can be modelled and predicted in terms of ecological systems, with shallow lake eutrophication as a central application. By introducing the key concepts in section 1 and applying them in section 2, we have illustrated how the fundamental ideas can be shown in a real world example. Additionally in section 1, carrying capacity was introduced to outline important notions of critical transitions that are not present in shallow lake eutrophication, for example logistic growth.

Shallow lake eutrophication perfectly represents a critical transition, with the key shift being between a clear and turbid state. By the representation of feedback systems, we have established that this change is irreversible. Early warning signals such as increased variance, increased autocorrelation and critical slowing down were introduced in the first section. Upon application to shallow lake eutrophication, we observed that the given EWSs were not equivalent to those given in the case study and our own mathematical modelling of a general differential equation for shallow lake eutrophication. This introduced us to the idea of limitations, meaning we should not fully depend on general EWSs to predict every critical transition. To combat this machine learning can be used, and examples of how machine learning have been used in the context of shallow lakes with references to nutrients were briefly introduced in section 2.3.3.

Overall, one key conclusion that can be made from this report is that even though early warning signals of critical transitions have been researched and have been backed up by many case studies, we cannot solely depend on them for predicting shifts. Especially in situations where the shift can be catastrophic, additional methods of prediction should be used to determine whether a tipping point is about to occur. One of these methods is machine learning and the use of surrogate data to make these predictions.

Appendix

A Variance and Autocorrelation Formulas

Variance is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2],$$

if the variance is high values are spread out from the mean, and if the variance is low values are clustered around the mean [100] [92]. The autocorrelation function is:

$$\rho(\tau) = \frac{\text{Cov}(X_t, X_{t+\tau})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+\tau})}}, \quad \text{Cov}(X_t, X_{t+\tau}) = \mathbb{E}[(X_t - \mu)(X_{t+\tau} - \mu)], \rho(\tau) \in [-1, 1]$$

If $\rho \approx 1$, there is a strong correlation at lag τ or if $\rho \approx 0$, there is little to no correlation at lag τ [107] [16].

B Fast-slow System Derivation

The following shows how we can use dynamical systems to form an ordinary differential equation (ODE), to mathematically show a critical transition. We do this by using the derivation from ‘A mathematical framework for critical transitions: Bifurcations, fast–slow systems and stochastic dynamics’ formulated by Christian Kuehn.

To start we will look at this parameterised family of ODEs, let $x \in \mathbb{R}^m$ represent phase space variables (where the phase space is all the possible states of a physical system [123]) and $y \in \mathbb{R}^m$ represent parameters:

$$\frac{dy}{dx} = x' = f(x; y) \tag{23}$$

A parameter evolves slowly until a tipping point is reached, so now we will include these parameters into the ODE:

$$\begin{aligned} x' &= f(x, y), \\ y' &= 0 \end{aligned} \tag{24}$$

Using (24), we will incorporate a slowly varying parameter by adding a slow evolution to y , introducing a small parameter ϵ , where $0 < \epsilon \ll 1$ and assuming a constant g to be sufficiently smooth. Generally, we can assume that $g \equiv 1$ when the parameter is decoupled from the dynamics of the phase space.

$$\begin{aligned} x' &= f(x, y), \\ y' &= \epsilon g(x, y) \end{aligned} \tag{25}$$

C Logistic Growth Model Derivation

To enhance the first logistic growth model given in section 1.2.4 and improve accuracy at early-stage population dynamics, we will look at a version of the logistic growth model that incorporates an exponential component. To find this model, we will use a derivation from ‘Analysis of Logistic Growth Models’ written by A. Tsoularis. To start we will look at the simplest realistic model of population dynamics, this is a model of exponential growth:

$$\frac{dN}{dt} = rN \quad (26)$$

This equation will give the solution:

$$N(t) = N_0 e^{rt} \quad (27)$$

Similarly to our first model, r represents the intrinsic growth rate (growth rate per capital) [127]. Now we want to eliminate unrestricted growth, Verhulst proposed that a stable population’s saturation level would be characteristic of the environment [134]. This caused the introduction of the exponential model, using a multiplicative factor: $1 - \frac{N}{K}$, representing the fractional deficiency of the current size from the saturation level, K . Within Lotka’s analysis of the logistic growth concept, it is stated that the rate of population growth $\frac{dN}{dt}$, at any moment t is a function of the population size at any moment, $N(t)$, therefore:

$$\frac{dN}{dt} = f(N) [82] \quad (28)$$

A zero population will have zero growth; therefore, $N=0$ is an algebraic root of the unknown function $f(N)$. We will now expand $f(N)$ as a Taylor series near $N=0$ and setting $f(0) = 0$, Lotka obtained this power series:

$$f(N) = N f'(0) + \frac{N^2}{2} f''(0) = N [f'(0) + \frac{N}{2} f''(0)] \quad (29)$$

In this case, higher terms are assumed to be negligible. Now, we set $f'(0) = r$ and $f''(0) = -\frac{2K}{r}$, where yet again, r is intrinsic growth rate and K is the carrying capacity. This leads to the Verhulst logistic equation or Verhulst-Pearl equation:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) \quad (30)$$

Verhulst first derived this as a curve, then Pearl [94] used the curve to approximate population growth. This equation gives the solution:

$$N(t) = \frac{K N_0}{(K - N_0)e^{-rt} + N_0} \quad (31)$$

Where N_0 represents the population size at time $t=0$ [127]. Both models seem almost identical, the main difference being that instead of A being a function of population density, we have a specific exponential function N , which takes into account the initial population size.

D Finding Bifurcations

To find a bifurcation, we first must be able to find a fixed point. A fixed point can be defined using the following definition from the book ‘Fixed Point Theory’ by Andrzej Granas and James Dugundji [46]:

Definition. Let $f : X \rightarrow X$ be a function from X to itself. A point $x \in X$ is called a fixed point if:

$$f(x) = x$$

To find a fixed point of an equation $f(x)$, we set $f(x) = 0$, and rearrange this equation to form a different equation $x = g(x)$ that has the same solution or root. The value $x = p$ where $p = g(p)$ is the fixed point [44]. Following this we can then form a Jacobian matrix using the original equation or equations. A Jacobian matrix can be defined in the following way, as by the definition given in the book ‘Chaos: an introduction to dynamical systems’ by Kathleen T. Alligood et al:

Definition. Let $\mathbf{f} = (f_1, f_2, \dots, f_m)$ be a map on $\mathbb{R}^m$ and let $\mathbf{p} \in \mathbb{R}^m$. The Jacobian matrix of $\mathbf{f}$ at $\mathbf{p}$, denoted $D\mathbf{f}(\mathbf{p})$, is the matrix:

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{p}) & \dots & \frac{\partial f_1}{\partial x_m}(\mathbf{p}) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{p}) & \dots & \frac{\partial f_m}{\partial x_m}(\mathbf{p}) \end{pmatrix}$$

of partial derivatives applied at $\mathbf{p}$.

Next we will find the eigenvalues that correspond to the jacobian matrix with the insertion of each fixed point. We do this by using the following formula:

$$\det(A - \lambda I) = 0 \quad (32)$$

Where A is our Jacobian matrix, λ is a scalar representing an eigenvalue and I is the identity matrix [101]. The eigenvalue will then help us classify the fixed points and bifurcation type.

We will now apply this method to an example:

Example. Consider the 2 dynamical equations below where $\dot{x} = \frac{dx}{dt}$ and $\dot{y} = \frac{dy}{dt}$ and t represents time:

$$\begin{aligned} \dot{x} &= x - y = f_1 \\ \dot{y} &= x^2 - 1 = f_2 \end{aligned} \quad (33)$$

To find the fixed points we set $\dot{x} = 0$ and $\dot{y} = 0$

$$\begin{aligned} x - y = 0 &\implies x = y \\ x^2 - 1 = (x - 1)(x + 1) = 0 &\implies x = 1, x = -1 \end{aligned}$$

Then inserting the values of x into $x = y$ we get the fixed points: $x_{*1} = (1, 1)$ and $x_{*2} = (-1, -1)$.

Now we form the Jacobian matrix and insert our fixed points into the matrix:

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2x & 0 \end{pmatrix} = Df$$

$$Df(x_{*1}) = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$$

$$Df(x_{*2}) = \begin{pmatrix} 1 & -1 \\ -2 & 0 \end{pmatrix}$$

Then we find the eigenvalues for $Df(x_{*1})$ and $Df(x_{*2})$:

$$Df(x_{*1}) = \begin{pmatrix} 1 - \lambda & -1 - \lambda \\ 2 - \lambda & 0 \end{pmatrix} = 0 - (-1 - \lambda)(2 - \lambda) = (1 + \lambda)(2 - \lambda) \therefore \lambda_{1,2} = -1, 2$$

$$Df(x_{*2}) = \begin{pmatrix} 1 - \lambda & -1 - \lambda \\ -2 - \lambda & 0 \end{pmatrix} = 0 - (-1 - \lambda)(-2 - \lambda) = (1 + \lambda)(2 + \lambda) \therefore \lambda_{3,4} = -1, -2$$

This gives the eigenvalues $\lambda_{1,2} = -1, 2$ and $\lambda_{3,4} = -1, -2$.

Using section 1.3.3 we can classify these fixed points $\lambda_{1,2} = -1, 2$ has one positive eigenvalue and one negative eigenvalue, so this is a saddle point. $\lambda_{3,4} = -1, -2$ are both negative, therefore represent a stable node.

E Derivation of Bifurcation Equations

E.1 Fold Bifurcation

We will now derive the normal form of a saddle-node bifurcation using the method given by Yuri A. Kuznetsov (2006):

We start off by considering a smooth ODE:

$$\dot{x} = f(x, \mu), x \in \mathbb{R} \quad (34)$$

Throughout this derivation, we assume that there is an equilibrium at $f(0, 0) = 0$. At a bifurcation point, the linear approximation will vanish:

$$\frac{\partial f}{\partial x}(0, 0) = 0$$

This insinuates that there is a zero eigenvalue, which is a key defining factor of a saddle-node bifurcation. To avoid any trivial cases we assume that:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &\neq 0 \\ \frac{\partial f}{\partial \mu} &\neq 0 \end{aligned}$$

Next, we perform a Taylor expansion around the point $(0, 0)$ [74], ignoring any higher-order terms, as they are much smaller than the leading terms and do not affect the qualitative behaviour due to them being close to the bifurcation point [120].

$$\dot{x} = a\mu + bx^2$$

Now, we rescale and let:

$$x = \alpha y$$

Specifically choosing α so the quadratic term will now be y^2 . Once we adjust the parameters, we can reduce this to:

$$\dot{y} = \mu + y^2$$

When putting this back into the normal form we get:

$$\dot{x} = \mu + x^2 \quad (35)$$

This is the normal form of the saddle node bifurcation. Upon solving the equilibrium conditions:

$$\mu + x^2 = 0 \Rightarrow x = \pm\sqrt{-\mu}$$

If $\mu < 0$, then there are two equilibria, one stable and one unstable, if $\mu = 0$ then there is a singular degenerate equilibrium and if $\mu > 0$ then there are no equilibria [74]. This explains the stated definition of the collision and disappearance of equilibria.

E.2 Hopf Bifurcation

We will now perform a derivation of the normal form of a hopf bifurcation, using steps from Strogatz and Yuri Kuznetsov.

First, we start with a non-linear system, similar to equation (16):

$$\dot{x} = f(x, \mu), x \in \mathbb{R}^2 \quad (36)$$

Yet again, we assume f is smooth, and has an equilibrium $f(0, \mu) = 0$. Then we compute the Jacobian:

$$A(\mu) = D_x f(0, \mu)$$

When near the equilibrium, this takes the form:

$$\dot{x} = A(\mu)x$$

Ignoring higher order terms for the same reasons given in the previous example [120]. Hopf bifurcations have purely imaginary eigenvalues, so we assume $\mu = 0$, therefore $\lambda_{1,2} = \pm i\omega_0$ where $\omega_0 \neq 0$. This infers that the equilibrium is neutrally stable [73].

Now, we introduce a condition to ensure that the eigenvalues cross the imaginary axis as μ varies:

$$\left. \frac{d}{d\mu} \Re(\lambda(\mu)) \right|_{\mu=0} \neq 0$$

Following this, we move to the complex coordinate form by letting $z = x_1 + ix_2$. This means that $\dot{z} = (\mu + i\omega_0)z + g(z, \bar{z}, \mu)$, where g is a non-linear term with an order greater than or equal to two [72].

We next perform a Taylor expansion, where only higher-order non-linearities appear:

$$g(z, \bar{z}, \mu) = a_{20}z^2 + a_{11}z\bar{z} + a_{02}\bar{z}^2 + a_{30}z^3 + a_{21}z^2\bar{z} + a_{12}z\bar{z}^2 + a_{03}\bar{z}^3 + \dots \quad [120]$$

Where a is referring to the coefficients of the Taylor expansion, and the subscripts (a_{ij}) are the power of the variable (e.g i is the power of x and j is the power of r [72]). The dynamics can be reduced to a 2D center manifold, as the eigenvalues are purely imaginary. Therefore $\dot{z} = F(z, \bar{z}, \mu)$. We can also remove non-resonant terms:

$$\dot{z} = (\mu + i\omega_0)z + cz|z|^2 \text{ where } c \in \mathbb{C} \quad [72]$$

Finally, we convert this into polar coordinates. Let $z = re^{i\theta}$, which we substitute into our previous equation and then separate into radial and angular dynamics.

Radial dynamics:

$$\dot{r} = r(\mu + \beta r^2) \text{ where } \beta = \Re(c)$$

Angular dynamics:

$$\dot{\theta} = \omega_0 + \mathcal{O}(r^2) \quad [120]$$

When we rescale the variables, we get the standard form:

$$\dot{r} = r(\mu - r^2) \quad [73] \quad (37)$$

If $\mu < 0$ then this is a stable fixed point, if $\mu = 0$ then the system is in critical oscillatory state and if $\mu > 0$ then within the system a stable limit cycle will emerge [120].

E.3 Pitchfork Bifurcation

Now we will perform a derivation of the normal form of a pitchfork bifurcation using steps from Strogatz, Kuznetsov and Guckenheimer.

Once again, we start by considering a dynamical system containing a singular parameter:

$$\dot{x} = f(x, r) \quad (38)$$

Where $x \in \mathbb{R}$ is a variable and $r \in \mathbb{R}$ is a parameter. We assume there is equilibrium at $f(0, 0) = 0$ [120]. Following this we will perform a Taylor expansion $f(x, r)$ about $(0, 0)$:

$$f(x, r) = a_{01}r + a_{10}x + a_{11}xr + a_{30}x^3 + \dots$$

This expansion shows the behaviour near the bifurcation point [72]. As $x = 0$ is an equilibrium for all the surrounding r , then $f(0, r) = 0$. This will eliminate constant terms (e.g $a_{01}r$), leaving:

$$f(x, r) = a_{10}x + a_{11}xr + a_{30}x^3 + \dots [47]$$

One property of pitchfork bifurcations is reflection symmetry, meaning $f(-x, r) = -f(x, r)$. Therefore this is an odd function, so we can remove all even powers:

$$f(x, r) = b_1(r)x + b_3x^3 + \dots$$

We have relabelled the Taylor-series coefficients for simplicity and to group powers of x [120]. Now let $b_1(r) = \mu r$, so the system now looks like:

$$\dot{x} = \mu r x + b_3 x^3 [72]$$

Now we rescale x and simply the coefficients to give the normal form:

$$\dot{x} = r x - x^3 [47] \quad (39)$$

E.4 Transcritical Bifurcation

Now we will derive the normal form of this bifurcation combining the methods of Strogatz, Kuznetsov and Guckenheimer:

We start by considering a dynamical system containing a singular parameter:

$$\dot{x} = f(x, r) \quad (40)$$

Which has an equilibrium at $f(0, r) = 0$ ensuring that $x = 0$ is always a fixed point [120]. Now we use a Taylor expansion around $(x, r) = (0, 0)$:

$$f(x, r) = a_{01}r + a_{10}x + a_{11}xr + a_{20}x^2 + \dots [72]$$

As $x = 0$ is an equilibrium for all r , then $f(0, r) = 0 \Rightarrow a_{01}r = 0$, so:

$$f(x, r) = a_{10}x + a_{11}xr + a_{20}x^2 + \dots [47]$$

At a transcritical bifurcation the linear term in x will vanish at the critical point $a_{10} = 0$, quadratic non-linearity will be shown $a_{20} \neq 0$, meaning:

$$f(x, r) = a_{11}xr + a_{20}x^2 + \dots [72]$$

Following this we take x as a factor to simplify:

$$f(x, r) = x(a_{11}r + a_{20}x)$$

This shows the two intersecting equilibria at $x = 0$ and $x = -\frac{a_{11}}{a_{20}}r$ [120]. Following this we scale x and t (as x is a function of t). $x \rightarrow 0$, $t \rightarrow \beta t$. Which simplifies the system to the normal form:

$$\dot{x} = rx - x^2 [47] \tag{41}$$

.

To better understand the bifurcation diagram we solve the equation:

$$\dot{x} = 0 \Rightarrow x(r - x) = 0$$

The solutions are $x = 0$ and $x = r$. This can be interpreted for different values of r , for example when $r < 0$ then $x = 0$ is stable, when $r > 0$ then $x = r$ is stable and $x = 0$ is unstable. These values of r and x can be seen by the coloured lines in figure 7.

F Establishing Equilibrium in Shallow Lake Eutrophication

Now we will use the method given appendix D to establish the equilibrium using equations 12, 14, 15 and 16. The equilibrium will take place when:

$$\frac{dE}{dt} = 0 = \frac{dV}{dt} \tag{42}$$

The fixed points therefore satisfy $E^* = E_0 \frac{h_e}{h_e + V^*}$ and $V^* = \frac{h_v^p}{h_v^p + E^{*p}}$. In a clear-water equilibrium there will be low E^* (turbidity) and high V^* , opposing the turbid equilibrium with high E^* and low V^* [108]. However, both states may coexist, forming an alternative stable equilibria. This is an unstable intermediate state which acts as a critical threshold. Therefore large perturbations can trigger unexpected regime shifts between the two ecological states [107] [111].

Now we redefine equations 25 and 27:

$$f(E, V) = r_e(E_0 \frac{h_e}{h_e + V} - E) \tag{43}$$

$$g(E, V) = r_v(\frac{h_v^p}{h_v^p + E^p} - V) \tag{44}$$

This gives the Jacobian matrix:

$$J(E, V) = \begin{pmatrix} \frac{\partial f}{\partial E} & \frac{\partial f}{\partial V} \\ \frac{\partial g}{\partial E} & \frac{\partial g}{\partial V} \end{pmatrix}$$

$$\begin{aligned}\frac{\partial f}{\partial E} &= -r_e \\ \frac{\partial f}{\partial V} &= -r_e E_0 \frac{h_e}{(h_e + V)^2} \\ \frac{\partial g}{\partial E} &= -r_v \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2} \\ \frac{\partial g}{\partial V} &= -r_v\end{aligned}$$

Therefore, the Jacobian matrix will look like:

$$J(E, V) = \begin{pmatrix} -r_e & -r_e E_0 \frac{h_e}{(h_e + V)^2} \\ -r_v \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2} & -r_v \end{pmatrix}$$

Now we calculate the determinant:

$$\det(J) = r_e r_v \left[1 - E_0 \frac{h_e}{(h_e + V)^2} \frac{p h_v^p E^{p-1}}{(h_v^p + E^p)^2} \right] \quad (45)$$

G R code for SDEs

```
##Parameters
E0 <- 4
he <- 1
hv <- 1
p <- 3

rv <- 1
re <- 1
sigma <- 0.1 # noise strength

dt <- 0.01 #time step size
T <- 200 #total time, this model is from t=0 to t=200
n <- as.integer(T/dt) + 1
#this is the number of steps needed

#Create 2 vectors of length n
E <- numeric(n)
V <- numeric(n)

E[1] <- 1 #at time t=0 the nutrient level =1
V[1] <- 2 #at time t=0, vegetation =2

for (i in 1:(n-1)) { #begins a loop

  dW <- rnorm(1, mean=0, sd=sqrt(dt)) #generates random increment, can be seen in brownian motion

  f <- re * (E0 * he/(he + V[i]) - E[i]) #defines the system
  g <- rv * (hv/(hv^p + E[i]^p) - V[i])

  E[i+1] <- E[i] + f*dt + sigma*dW #our second set of adjusted equations with modelling assumption
  V[i+1] <- V[i] + g*dt
}
```



```

time <- seq(0, T, by=dt) #time vector

plot(time, E, type="l", xlab="Time", ylab="Turbidity (E)",
      main="Stochastic Shallow Lake Dynamics")

window_size <- 500

#Rolling variance
var_roll <- rollapply(E, width=window_size, FUN=var, fill=NA)

#Rolling lag-1 autocorrelation
acf1 <- function(x) {
  acf(x, lag.max=1, plot=FALSE)$acf[2]
}

acf_roll <- rollapply(E, width=window_size, FUN=acf1, fill=NA)

par(mfrow=c(1,1))

#Variance
plot(time, var_roll, type="l", main="Rolling Variance")

#Autocorrelation
plot(time, acf_roll, type="l", main="Rolling Autocorrelation (lag-1)")

```

References

- [1] Zafar Ahsan. *Differential equations and their applications*. PHI Learning Pvt. Ltd., 2016.
- [2] Vladimir I Arnold. Lectures on bifurcations in versal families. In *Vladimir I. Arnold-Collected Works: Hydrodynamics, Bifurcation Theory, and Algebraic Geometry 1965-1972*, pages 271–340. Springer, 2013.
- [3] Peter Ashwin, Sebastian Wieczorek, Renato Vitolo, and Peter Cox. Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 370(1662):1166–1184, 2012.
- [4] John W Barko. Sediment interactions with submersed macrophytes. *CLAMERNOTli*, page 87, 1991.
- [5] Julius S. Bendat and Allan G. Piersol. *Random Data: Analysis and Measurement Procedures*. Wiley, 4th edition, 2010.
- [6] Elisa Benincà, Jef Huisman, Reinhard Heerkloss, Klaus D Jöhnk, Pedro Branco, Egbert H Van Nes, Marten Scheffer, and Stephen P Ellner. Chaos in a long-term experiment with a plankton community. *Nature*, 451(7180):822–825, 2008.
- [7] Pierre Bergé, Yves Pomeau, and Christian Vidal. *Order within chaos*, 1986.
- [8] Reinette Biggs, Stephen R Carpenter, and William A Brock. Turning back from the brink: detecting an impending regime shift in time to avert it. *Proceedings of the National academy of Sciences*, 106(3):826–831, 2009.
- [9] L. C. Birch. The intrinsic rate of natural increase: an historical perspective. *Annals of Botany*, 83(6):713–723, 1999.
- [10] Simon Birkholz, Carsten Brée, Ayhan Demircan, and Günter Steinmeyer. Predictability of rogue events. *Physical review letters*, 114(21):213901, 2015.
- [11] Nils Bochow, Anna Poltronieri, Martin Rypdal, and Niklas Boers. Reconstructing historical climate fields with deep learning. *Science Advances*, 11(14):eadp0558, 2025.
- [12] Carl Boettiger and Alan Hastings. Quantifying limits to detection of early warning for critical transitions. *Journal of the Royal Society Interface*, 9(75):2527–2539, 2012.
- [13] Carl Boettiger, Noam Ross, and Alan Hastings. Early warning signals: the charted and uncharted territories. *Theoretical ecology*, 6(3):255–264, 2013.
- [14] Jarrett Booz, Wei Yu, Guobin Xu, David Griffith, and Nada Golmie. A deep learning-based weather forecast system for data volume and recency analysis. In *2019 international conference on computing, networking and communications (ICNC)*, pages 697–701. IEEE, 2019.
- [15] George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. *Time Series Analysis: Forecasting and Control*. Wiley, 5th edition, 2015.
- [16] George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. *Time Series Analysis: Forecasting and Control*. Wiley, 5th edition, 2016.
- [17] William E. Boyce and Richard C. DiPrima. *Elementary Differential Equations and Boundary Value Problems*. John Wiley & Sons, Hoboken, NJ, 10th edition, 2012.
- [18] Martin Braun and Martin Golubitsky. *Differential equations and their applications*, volume 4. Springer, 1983.
- [19] William A Brock and Stephen R Carpenter. Interacting regime shifts in ecosystems: implication for early warnings. *Ecological Monographs*, 80(3):353–367, 2010.

- [20] Vince L Butitta, Stephen R Carpenter, Luke C Loken, Michael L Pace, and Emily H Stanley. Spatial early warning signals in a lake manipulation. *Ecosphere*, 8(10):e01941, 2017.
- [21] Jennifer Carpenter. May the best analyst win, 2011.
- [22] Stephen R Carpenter and William A Brock. Rising variance: a leading indicator of ecological transition. *Ecology letters*, 9(3):311–318, 2006.
- [23] Stephen R Carpenter, Donald Ludwig, and William A Brock. Management of eutrophication for lakes subject to potentially irreversible change. *Ecological applications*, 9(3):751–771, 1999.
- [24] Jyoti Champanerkar and Denis Blackmore. Pitchfork bifurcations of invariant manifolds. *Topology and its Applications*, 154(8):1650–1663, 2007.
- [25] Pascal Chossat and Reiner Lauterbach. *Methods in equivariant bifurcations and dynamical systems*, volume 15. World Scientific Publishing Company, 2000.
- [26] Christopher F Clements and Arpat Ozgul. Rate of forcing and the forecastability of critical transitions. *Ecology and Evolution*, 6(21):7787–7793, 2016.
- [27] Pierre Collet and J-P Eckmann. *Iterated maps on the interval as dynamical systems*. Springer Science & Business Media, 2009.
- [28] Carlos Corvalan, Simon Hales, and Anthony J McMichael. *Ecosystems and human well-being: health synthesis*. World Health Organization, 2005.
- [29] Eduardo Cotilla-Sanchez, Paul DH Hines, and Christopher M Danforth. Predicting critical transitions from time series synchrophasor data. *IEEE Transactions on smart grid*, 3(4):1832–1840, 2012.
- [30] Vasilis Dakos, Stephen R Carpenter, William A Brock, Aaron M Ellison, Vishwesha Guttal, Anthony R Ives, Sonia Kéfi, Valerie Livina, David A Seekell, Egbert H van Nes, et al. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PloS one*, 7(7):e41010, 2012.
- [31] Vasilis Dakos, Stephen R Carpenter, Egbert H van Nes, and Marten Scheffer. Resilience indicators: prospects and limitations for early warnings of regime shifts. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1659), 2015.
- [32] Vasilis Dakos, Egbert H Van Nes, Paolo d’Odorico, and Marten Scheffer. Robustness of variance and autocorrelation as indicators of critical slowing down. *Ecology*, 93(2):264–271, 2012.
- [33] Smita Deb, Sahil Sidheekh, Christopher F Clements, Narayanan C Krishnan, and Partha S Dutta. Machine learning methods trained on simple models can predict critical transitions in complex natural systems. *Royal Society Open Science*, 9(2), 2022.
- [34] Pablo Del Monte-Luna, Barry W Brook, Manuel J Zetina-Rejón, and Victor H Cruz-Escalona. The carrying capacity of ecosystems. *Global ecology and biogeography*, 13(6):485–495, 2004.
- [35] Peter D Ditlevsen and Sigfus J Johnsen. Tipping points: early warning and wishful thinking. *Geophysical research letters*, 37(19), 2010.
- [36] John M Drake. Early warning signals of stochastic switching. *Proceedings of the Royal Society B: Biological Sciences*, 280(1766), 2013.
- [37] J. Duarte et al. On chaos, tipping and delayed dynamical transitions in a hassell-type population model with an allee effect. *Mathematics*, 13(8):1275, 2025.
- [38] Veit Elser, I Rankenburg, and P Thibault. Searching with iterated maps. *Proceedings of the National Academy of Sciences*, 104(2):418–423, 2007.

- [39] Paul G. Falkowski and John A. Raven. *Aquatic Photosynthesis*. Princeton University Press, Princeton, NJ, 2007.
- [40] Brian D Fath, Sven E Jørgensen, Bernard C Patten, and Milan Straškraba. Ecosystem growth and development. *Biosystems*, 77(1-3):213–228, 2004.
- [41] L. Fung and Y. Hwang. A sequence of transcritical bifurcations in a suspension of gyrotactic microswimmers in a vertical pipe. *Journal of Fluid Mechanics*, 902:R2, 2020.
- [42] Y. Gao et al. A bayesian neural network model for chlorophyll-a prediction in lake taihu, china. *Environmental Engineering Science*, 32(10), 2015.
- [43] Periklis Gogas and Theophilos Papadimitriou. Machine learning in economics and finance. *Computational Economics*, 57(1):1–4, 2021.
- [44] Richard B. Goldstein. Fixed points / iterations, 2023. Lecture notes on numerical analysis.
- [45] Daniel Graeber, Mark J McCarthy, Tom Shatwell, Dietrich Borchardt, Erik Jeppesen, Martin Søndergaard, Torben L Lauridsen, and Thomas A Davidson. Consistent stoichiometric long-term relationships between nutrients and chlorophyll-a across shallow lakes. *Nature Communications*, 15(1):809, 2024.
- [46] Andrzej Granas and James Dugundji. *Fixed Point Theory*. Springer, New York, 2003.
- [47] John Guckenheimer and Philip Holmes. *Nonlinear oscillations, dynamical systems, and bifurcations of vector fields*. Springer Science & Business Media, 2013.
- [48] Amir Hamzeh Haghiabi, Ali Heidar Nasrolahi, and Abbas Parsaie. Water quality prediction using machine learning methods. *Water Quality Research Journal*, 53(1):3–13, 2018.
- [49] J Michael Harrison. *Brownian models of performance and control*. Cambridge University Press, 2013.
- [50] Alan Hastings and Derin B Wysham. Regime shifts in ecological systems can occur with no warning. *Ecology letters*, 13(4):464–472, 2010.
- [51] Hermann Held and Thomas Kleinen. Detection of climate system bifurcations by degenerate fingerprinting. *Geophysical Research Letters*, 31(23), 2004.
- [52] James PJ Hetherington, Anne Warner, and Robert M Seymour. Simplification and its consequences in biological modelling: conclusions from a study of calcium oscillations in hepatocytes. *Journal of the Royal Society Interface*, 3(7):319–331, 2006.
- [53] Marina Hirota, Milena Holmgren, Egbert H Van Nes, and Marten Scheffer. Global resilience of tropical forest and savanna to critical transitions. *Science*, 334(6053):232–235, 2011.
- [54] SH Hosper. Biomanipulation, new perspectives for restoration of shallow, eutrophic lakes in the netherlands. *Hydrobiological Bulletin*, 23(1):5–10, 1989.
- [55] Y. Huang, Z. Zhang, et al. Prediction of chlorophyll-a and risk assessment of water blooms in poyang lake based on machine learning methods. *Environmental Pollution*, 2024.
- [56] IBM. What is a time series model?, 2026. Accessed: 2026-04-26.
- [57] Hamza Ahmad Isiyaka, Adamu Mustapha, Hafizan Juahir, and Philip Phil-Eze. Water quality modelling using artificial neural network and multivariate statistical techniques. *Modeling Earth Systems and Environment*, 5(2):583–593, 2019.
- [58] Anthony R Ives. Diversity and stability in ecological. *Theoretical ecology: principles and applications*, page 98, 2007.
- [59] Jan H Janse. A model of nutrient dynamics in shallow lakes in relation to multiple stable states. *Hydrobiologia*, 342(0):1–8, 1997.

- [60] Jan H Janse, Lisette N De Senerpont Domis, Marten Scheffer, Lambertus Lijklema, Lowie Van Liere, Marcel Klinge, and Wolf M Mooij. Critical phosphorus loading of different types of shallow lakes and the consequences for management estimated with the ecosystem model pclake. *Limnologia*, 38(3-4):203–219, 2008.
- [61] JH Janse, M Scheffer, L Lijklema, L Van Liere, JS Sloot, and WM Mooij. Estimating the critical phosphorus loading of shallow lakes with the ecosystem model pclake: sensitivity, calibration and uncertainty. *Ecological Modelling*, 221(4):654–665, 2010.
- [62] E Jeppesen, JP Jensen, P Kristensen, Ma Søndergaard, E Mortensen, O Sortkjaer, and K Olrik. Fish manipulation as a lake restoration tool in shallow, eutrophic, temperate lakes 2: threshold levels, long-term stability and conclusions. *Hydrobiologia*, 200(1):219–227, 1990.
- [63] Erik Jeppesen, Martin Søndergaard, Brian Kronvang, Jens P Jensen, Lars M Svendsen, and Torben L Lauridsen. Lake and catchment management in denmark. *Hydrobiologia*, 395(0):419–432, 1999.
- [64] Erik Jeppesen, Martin Søndergaard, Morten Søndergaard, and Kirsten Christoffersen. *The structuring role of submerged macrophytes in lakes*, volume 131. Springer Science & Business Media, 2012.
- [65] WR Johns. Computer-aided chemical engineering. *Kirk-Othmer Encyclopedia of Chemical Technology*, 2000.
- [66] K. Jyothi. A study on differential equations (exactness). *International Journal of Novel Research and Development*, 2023. Available online.
- [67] John Kambhu, Neel Krishnan, and Scott Weidman. New directions for understanding systemic risk: a report on a conference cosponsored by the federal reserve bank of new york and the national academy of sciences. *Economic Policy Review*, 13(Nov), 2007.
- [68] Sonia Kéfi, Vishwesha Guttal, William A Brock, Stephen R Carpenter, Aaron M Ellison, Valerie N Livina, David A Seekell, Marten Scheffer, Egbert H Van Nes, and Vasilis Dakos. Early warning signals of ecological transitions: methods for spatial patterns. *PloS one*, 9(3):e92097, 2014.
- [69] Thomas Kleinen, Hermann Held, and Gerhard Petschel-Held. The potential role of spectral properties in detecting thresholds in the earth system: application to the thermohaline circulation. *Ocean Dynamics*, 53(2):53–63, 2003.
- [70] Ling-Wei Kong, Hua-Wei Fan, Celso Grebogi, and Ying-Cheng Lai. Machine learning prediction of critical transition and system collapse. *Physical Review Research*, 3(1):013090, 2021.
- [71] Christian Kuehn. A mathematical framework for critical transitions: Bifurcations, fast–slow systems and stochastic dynamics. *Physica D: Nonlinear Phenomena*, 240(12):1020–1035, 2011.
- [72] Yuri A Kuznetsov. *Elements of applied bifurcation theory*. Springer, 1998.
- [73] Yuri A. Kuznetsov. Hopf bifurcation. *Scholarpedia*, 1(10):1857, 2006.
- [74] Yuri A Kuznetsov. Saddle-node bifurcation. *Scholarpedia*, 1(10):1859, 2006.
- [75] Gemma Lancaster, Dmytro Iatsenko, Aleksandra Pidde, Valentina Ticcinelli, and Aneta Stefanovska. Surrogate data for hypothesis testing of physical systems. *Physics reports*, 748:1–60, 2018.
- [76] Timothy M Lenton, Hermann Held, Elmar Kriegler, Jim W Hall, Wolfgang Lucht, Stefan Rahmstorf, and Hans Joachim Schellnhuber. Tipping elements in the earth’s climate system. *Proceedings of the national Academy of Sciences*, 105(6):1786–1793, 2008.
- [77] Timothy M Lenton, VN Livina, Vasilis Dakos, Egbert H van Nes, and M Scheffer. Early warning of climate tipping points from critical slowing down: comparing methods to improve robustness. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 370(1962):1185–1204, 2012.

- [78] Diya Li and Jianxi Gao. Towards perturbation prediction of biological networks using deep learning. *Scientific reports*, 9(1):11941, 2019.
- [79] Zhenwei Li, James R Kermode, and Alessandro De Vita. Molecular dynamics with on-the-fly machine learning of quantum-mechanical forces. *Physical review letters*, 114(9):096405, 2015.
- [80] Valerie N Livina, F Kwasniok, and Timothy M Lenton. Potential analysis reveals changing number of climate states during the last 60 kyr. *Climate of the Past*, 6(1):77–82, 2010.
- [81] VN Livina, PD Ditlevsen, and TM Lenton. An independent test of methods of detecting system states and bifurcations in time-series data. *Physica A: Statistical Mechanics and its Applications*, 391(3):485–496, 2012.
- [82] Alfred J Lotka. Elements of mathematical biology. *New York: Oxford University press*, page P158, 1956.
- [83] Zhiqin Ma, Chunhua Zeng, Yi-Cheng Zhang, and Thomas M Bury. Learning from the past: predicting critical transitions with machine learning trained on surrogates of historical data. *arXiv preprint arXiv:2410.09707*, 2024.
- [84] Zhiqin Ma, Chunhua Zeng, Yi-Cheng Zhang, and Thomas M Bury. Predicting critical transitions with machine learning trained on surrogates of historical data. *Communications Physics*, 8(1):258, 2025.
- [85] Holger R Maier and Graeme C Dandy. Neural networks for the prediction and forecasting of water resources variables: a review of modelling issues and applications. *Environmental modelling & software*, 15(1):101–124, 2000.
- [86] Robert M. May. Simple mathematical models with very complicated dynamics. *Nature*, 261:459–467, 1976.
- [87] Robert M May. Thresholds and breakpoints in ecosystems with a multiplicity of stable states. *Nature*, 269(5628):471–477, 1977.
- [88] Robert M May, Simon A Levin, and George Sugihara. Ecology for bankers. *Nature*, 451(7181):893–894, 2008.
- [89] Axel Meyer. Phylogenetic relationships and evolutionary processes in east african cichlid fishes. *Trends in ecology & evolution*, 8(8):279–284, 1993.
- [90] Jacques Monod. The growth of bacterial cultures. *Annual Review of Microbiology*, 3:371–394, 1949.
- [91] Crispin M Mutshinda and Robert B O’Hara. On the setting of environmental noise and the performance of population dynamical models. *BMC ecology*, 10(1):7, 2010.
- [92] Athanasios Papoulis and S. Unnikrishna Pillai. *Probability, Random Variables, and Stochastic Processes*. McGraw-Hill, 4th edition, 2002.
- [93] Jaideep Pathak, Brian Hunt, Michelle Girvan, Zhixin Lu, and Edward Ott. Model-free prediction of large spatiotemporally chaotic systems from data: A reservoir computing approach. *Physical review letters*, 120(2):024102, 2018.
- [94] Raymond Pearl and Lowell J Reed. On the rate of growth of the population of the united states since 1790 and its mathematical representation. *Proceedings of the national academy of sciences*, 6(6):275–288, 1920.
- [95] Lawrence Perko. *Differential equations and dynamical systems*, volume 7. Springer Science & Business Media, 2013.
- [96] Daniele Proverbio, Alexander Skupin, and Jorge Gonçalves. Systematic analysis and optimization of early warning signals for critical transitions using distribution data. *Isience*, 26(7), 2023.

- [97] Markus Reichstein, Gustau Camps-Valls, Bjorn Stevens, Martin Jung, Joachim Denzler, Nuno Carvalhais, and F Prabhat. Deep learning and process understanding for data-driven earth system science. *Nature*, 566(7743):195–204, 2019.
- [98] Colin S. Reynolds. *The Ecology of Phytoplankton*. Cambridge University Press, Cambridge, 2006.
- [99] Elizabeth Rohde, Nolan JT Pearce, Joelle Young, and Marguerite A Xenopoulos. Applying early warning indicators to predict critical transitions in a lake undergoing multiple changes. *Ecological Applications*, 32(7):e2685, 2022.
- [100] Sheldon M. Ross. *Introduction to Probability Models*. Academic Press, 11th edition, 2014.
- [101] Alistair Savage. Linear algebra i, 2018.
- [102] Marten Scheffer. Multiplicity of stable states in freshwater systems. *Hydrobiologia*, 200(1):475–486, 1990.
- [103] Marten Scheffer. *Ecology of shallow lakes*, volume 1. Springer, 1998.
- [104] Marten Scheffer. *Critical transitions in nature and society*. Princeton University Press, 2009.
- [105] Marten Scheffer, Jordi Bascompte, William A Brock, Victor Brovkin, Stephen R Carpenter, Vasilis Dakos, Hermann Held, Egbert H Van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009.
- [106] Marten Scheffer, Stephen R Carpenter, Timothy M Lenton, Jordi Bascompte, William Brock, Vasilis Dakos, Johan Van de Koppel, Ingrid A Van de Leemput, Simon A Levin, Egbert H Van Nes, et al. Anticipating critical transitions. *science*, 338(6105):344–348, 2012.
- [107] Marten Scheffer, Steve Carpenter, Jonathan A Foley, Carl Folke, and Brian Walker. Catastrophic shifts in ecosystems. *Nature*, 413(6856):591–596, 2001.
- [108] Marten Scheffer, S Harry Hosper, Marie Louise Meijer, Brian Moss, and Erik Jeppesen. Alternative equilibria in shallow lakes. *Trends in ecology & evolution*, 8(8):275–279, 1993.
- [109] Marten Scheffer and Erik Jeppesen. Regime shifts in shallow lakes. *Ecosystems*, 10(1):1–3, 2007.
- [110] Marten Scheffer, Sergio Rinaldi, Alessandra Gagnani, Luuc R Mur, and Egbert H van Nes. On the dominance of filamentous cyanobacteria in shallow, turbid lakes. *Ecology*, 78(1):272–282, 1997.
- [111] Marten Scheffer and Egbert H van Nes. Shallow lakes theory revisited: various alternative regimes driven by climate, nutrients, depth and lake size. *Hydrobiologia*, 584(1):455–466, 2007.
- [112] David W Schindler. Evolution of phosphorus limitation in lakes: natural mechanisms compensate for deficiencies of nitrogen and carbon in eutrophied lakes. *Science*, 195(4275):260–262, 1977.
- [113] Sebastian Schreiber and Volker HW Rudolf. Crossing habitat boundaries: coupling dynamics of ecosystems through complex life cycles. *Ecology letters*, 11(6):576–587, 2008.
- [114] Sebastian J Schreiber. Allee effects, extinctions, and chaotic transients in simple population models. *Theoretical population biology*, 64(2):201–209, 2003.
- [115] Thomas Schreiber and Andreas Schmitz. Improved surrogate data for nonlinearity tests. *Physical review letters*, 77(4):635, 1996.
- [116] Carl P Simon, Lawrence Blume, et al. *Mathematics for economists*, volume 7. Norton New York, 1994.
- [117] Hal L. Smith and Paul Waltman. *Monod Kinetics*. Chapman and Hall, 1995.
- [118] Val H. Smith. Eutrophication of freshwater and coastal marine ecosystems: a global problem. *Environmental Science and Pollution Research*, 10(2):126–139, 2003.

- [119] DHN Spence. Studies on the vegetation of shetland: Ii. reasons for the restriction of the exclusive pioneers to serpentine debris. *The Journal of Ecology*, pages 641–649, 1959.
- [120] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2 edition, 2018.
- [121] X. Sun et al. Machine learning models for chlorophyll-a forecasting in a freshwater lake: Case study of lake taihu. *Water*, 17(8), 2025.
- [122] Lang Tan, Zongzhi Wang, Ying Bai, and Xiaorong Huang. Short-term responses of nutrients and algal biomass in a eutrophic shallow lake to different scales of water transfer. *Science of The Total Environment*, 880:163321, 2023.
- [123] Terence Tao. Phase space. https://www.math.ucla.edu/~tao/preprints/phase_space.pdf, 2023. *Lecture notes, UCLA*.
- [124] James Theiler, Stephen Eubank, André Longtin, Bryan Galdrikian, and J Doynne Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D: Nonlinear Phenomena*, 58(1-4):77–94, 1992.
- [125] John Michael Tutill Thompson and H Bruce Stewart. *Nonlinear dynamics and chaos*. John Wiley & Sons, 2002.
- [126] Charles Tresser, Pierre Couillet, and Edson de Faria. Period doubling. *Scholarpedia*, 9(6):3958, 2014.
- [127] Anastasios Tsoularis and James Wallace. Analysis of logistic growth models. *Mathematical biosciences*, 179(1):21–55, 2002.
- [128] E Van Donk, MP Grimm, RD Gulati, and JPG Klein Breteler. Whole-lake food-web manipulation as a means to study community interactions in a small ecosystem. *Hydrobiologia*, 200(1):275–289, 1990.
- [129] Nicolaas G Van Kampen. Stochastic differential equations. *Physics reports*, 24(3):171–228, 1976.
- [130] Egbert H van Nes and Marten Scheffer. Large species shifts triggered by small forces. *The American Naturalist*, 164(2):255–266, 2004.
- [131] Dianneke van Wijk, Manqi Chang, Annette BG Janssen, Sven Teurlincx, and Wolf M Mooij. Regime shifts in shallow lakes explained by critical turbidity. *Water Research*, 242:119950, 2023.
- [132] Anna Vanselow, Sebastian Wieczorek, and Ulrike Feudel. When very slow is too fast – collapse of a predator–prey system. *Journal of Theoretical Biology*, 479:64–72, 2019.
- [133] Jose G Venegas, Tilo Winkler, Guido Musch, Marcos F Vidal Melo, Dominick Layfield, Nora Tgavalekos, Alan J Fischman, Ronald J Callahan, Giacomo Bellani, and R Scott Harris. Self-organized patchiness in asthma as a prelude to catastrophic shifts. *Nature*, 434(7034):777–782, 2005.
- [134] Pierre-François Verhulst. Notice sur la loi que la population suit dans son accroissement. *Correspondence mathématique et physique*, 10:113–129, 1838.
- [135] Paula Villa Martín, Juan A Bonachela, Simon A Levin, and Miguel A Muñoz. Eluding catastrophic shifts. *Proceedings of the National Academy of Sciences*, 112(15):E1828–E1836, 2015.
- [136] Rong Wang, John A Dearing, Peter G Langdon, Enlou Zhang, Xiangdong Yang, Vasilis Dakos, and Marten Scheffer. Flickering gives early warning signals of a critical transition to a eutrophic lake state. *Nature*, 492(7429):419–422, 2012.
- [137] Robert G. Wetzel. *Limnology: Lake and River Ecosystems*. Academic Press, San Diego, CA, 3rd edition, 2001.
- [138] Daniel S. Wilks. *Statistical Methods in the Atmospheric Sciences*. Elsevier, Amsterdam, 4th edition, 2019.

- [139] Han Yan, Feng Zhang, and Jin Wang. Thermodynamic and dynamical predictions for bifurcations and non-equilibrium phase transitions. *Communications Physics*, 6(1):110, 2023.
- [140] Jian Zhou et al. Why are shallow lakes prone to eutrophication?, 2022.

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